# Lab Exercise 2: **Deploy an Operating System on a Type 2 Hypervisor**

**Goal:** In this lab you will install a virtual machine on a type 2 hypervisor.

**Lab high-level steps:**

1. Create a virtual machine (VM) on the hypervisor from an **Ubuntu Desktop** ISO image.
2. Setup and configure the virtual machine operating system (Ubuntu Desktop 26.04.1 LTS).
3. Verify VMware Tools is installed on the Ubuntu Desktop VM.

**Before you start:**

- Be sure you have downloaded the correct **Ubuntu Desktop** installation file (**.iso**) to your laptop.
  This will be:
  - ubuntu-26.04.1-desktop-arm64.iso [for macOS with M1, M2, M3, M4, M5 CPUs]
  - ubuntu-26.04.1-desktop-amd64.iso [for Windows with Intel or AMD CPUs]

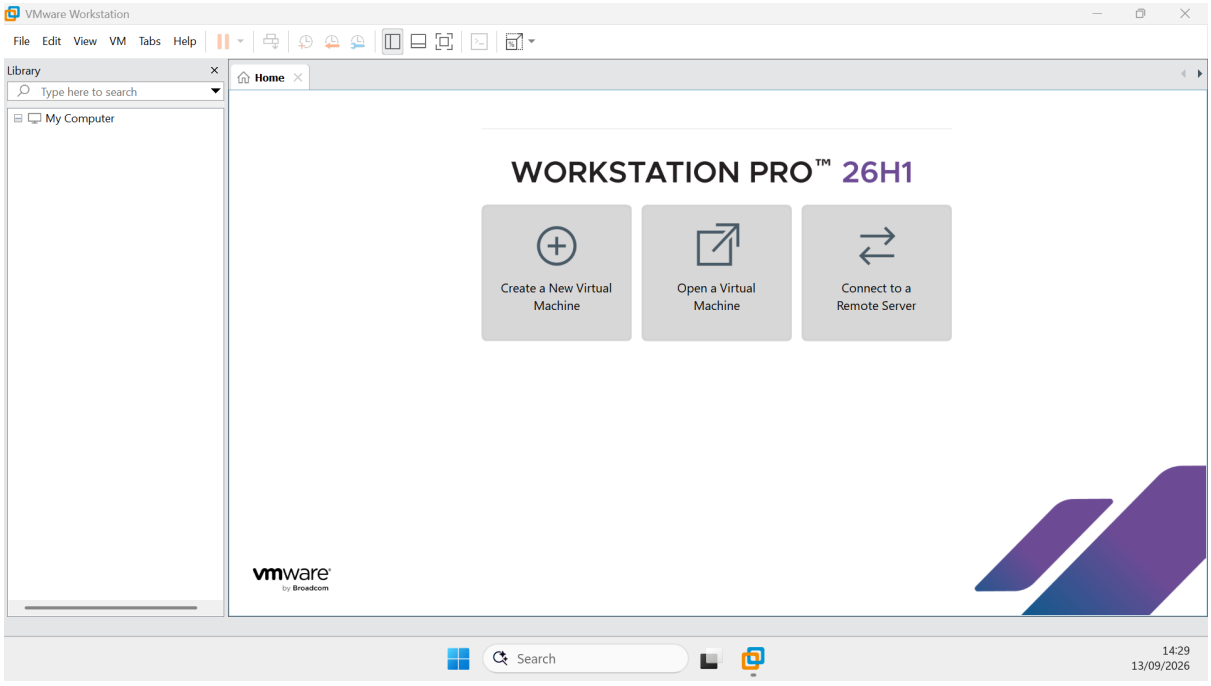
**Copy** the Ubuntu ISO file from the USB drive handed out in class. Do not run the installer from the USB drive!

# 1. Install Ubuntu Desktop

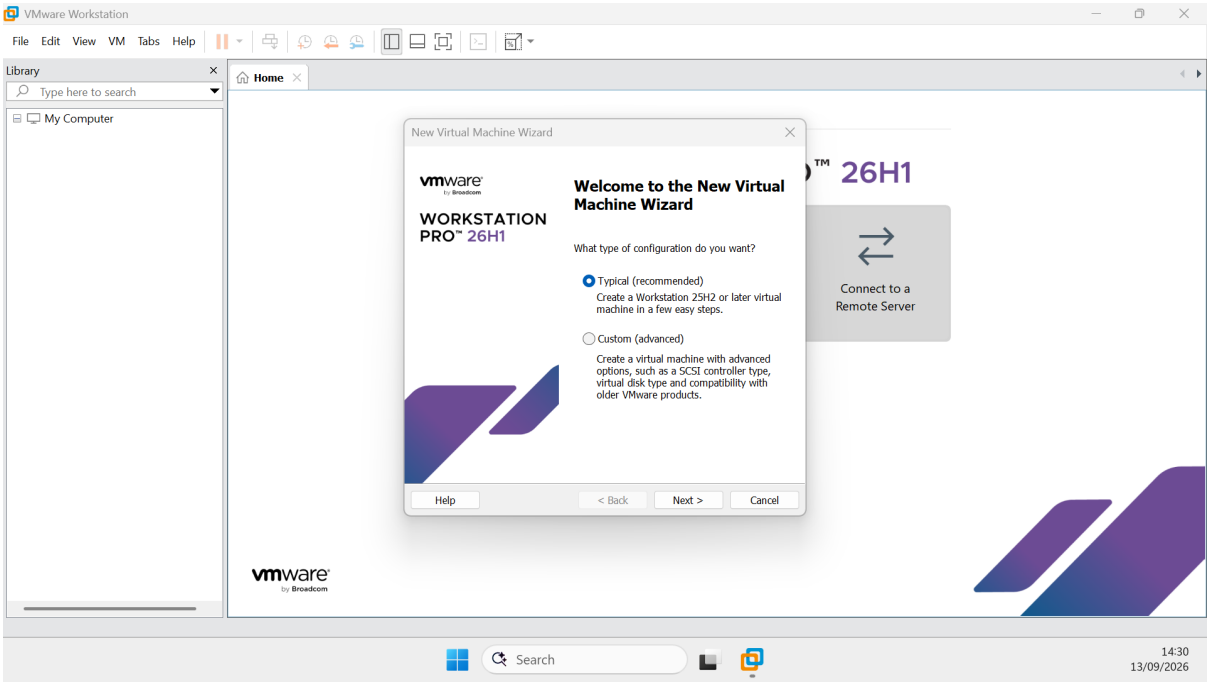
1.1. Open the **VMWare Workstation** application.



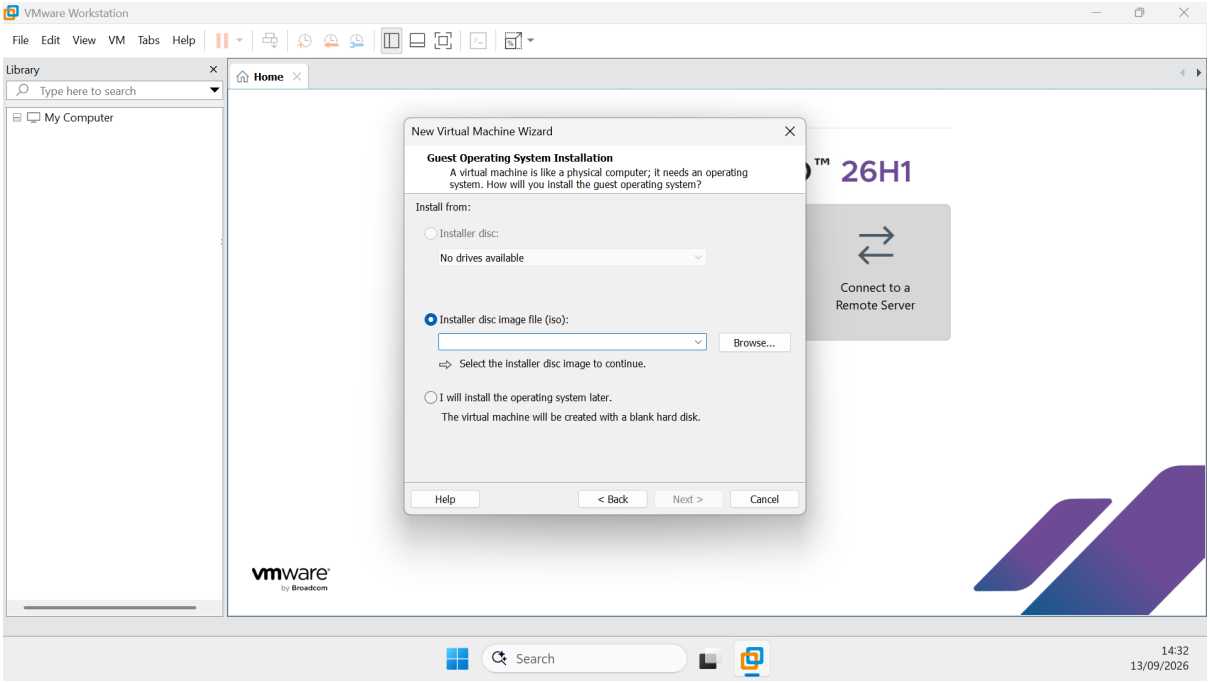

1.2. On the VMWare Workstation Home page, click the **Create a New Virtual Machine** option.



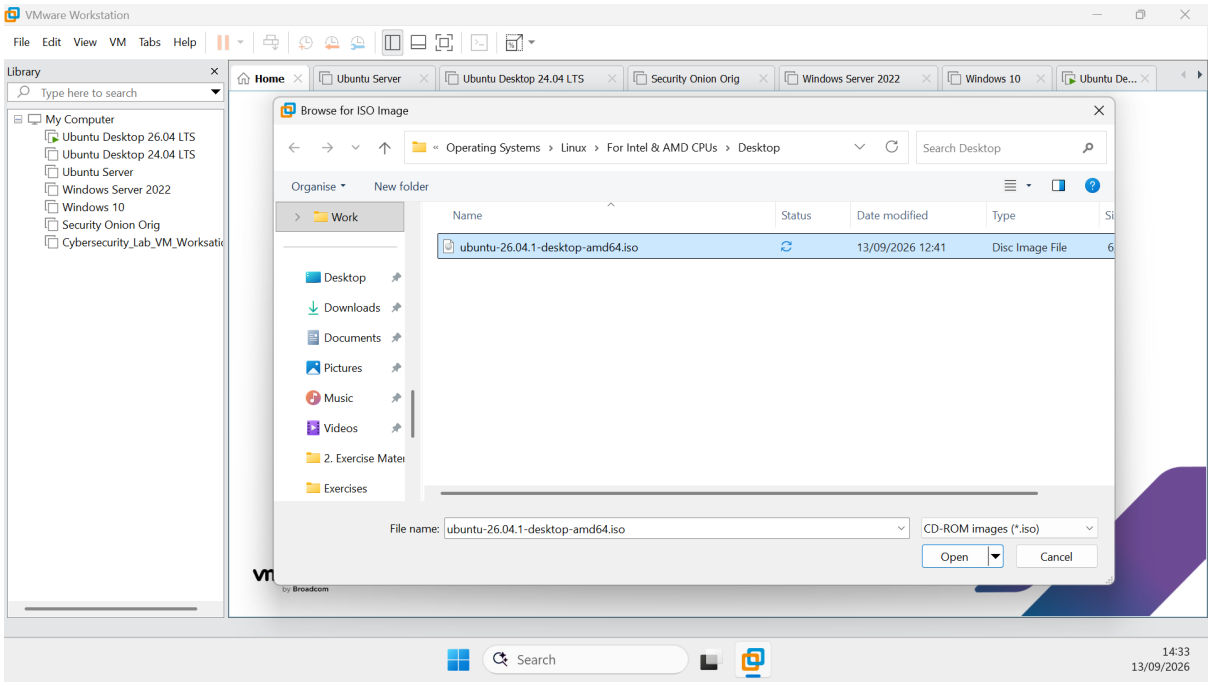
1.3. Select the **Typical (recommended)** option and click **Next**.



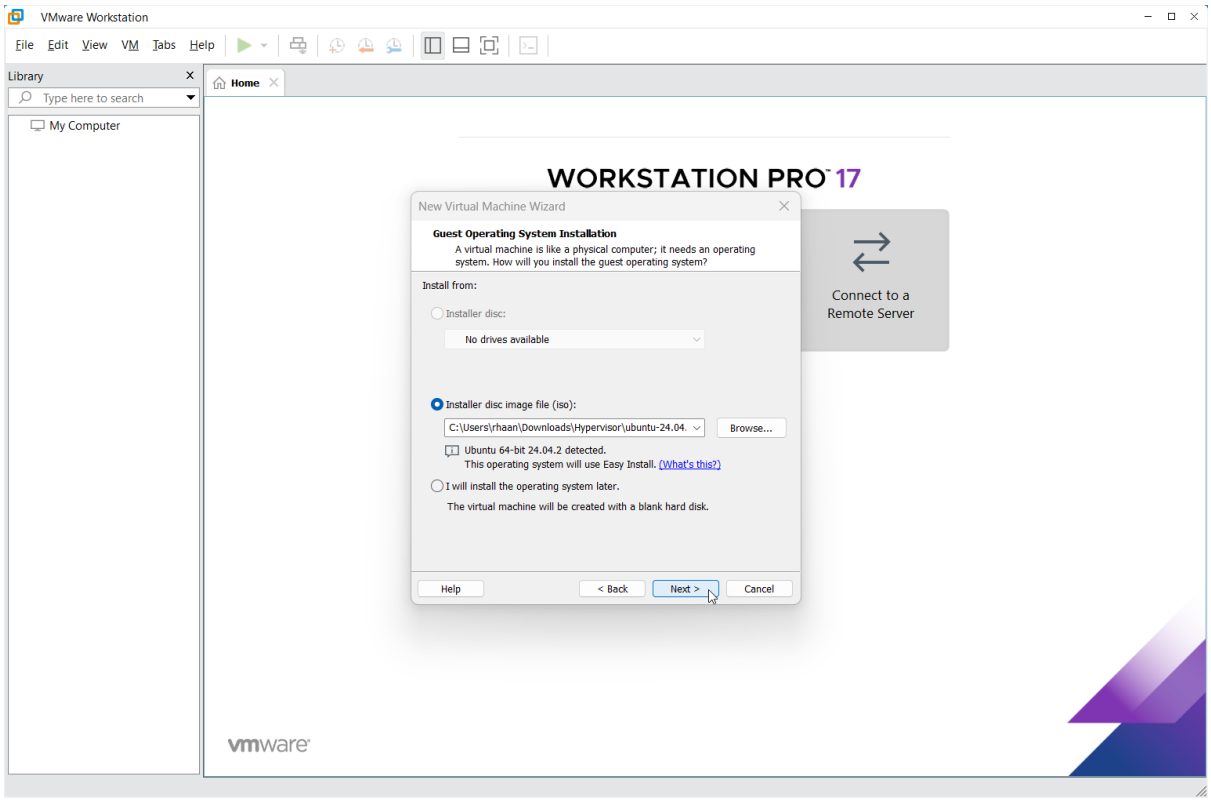

1.4. Select the Installer disc image (iso) option and click **Browse**.




1.5. Navigate to where you downloaded the Ubuntu Desktop ISO file. Select the file and click **Open**.

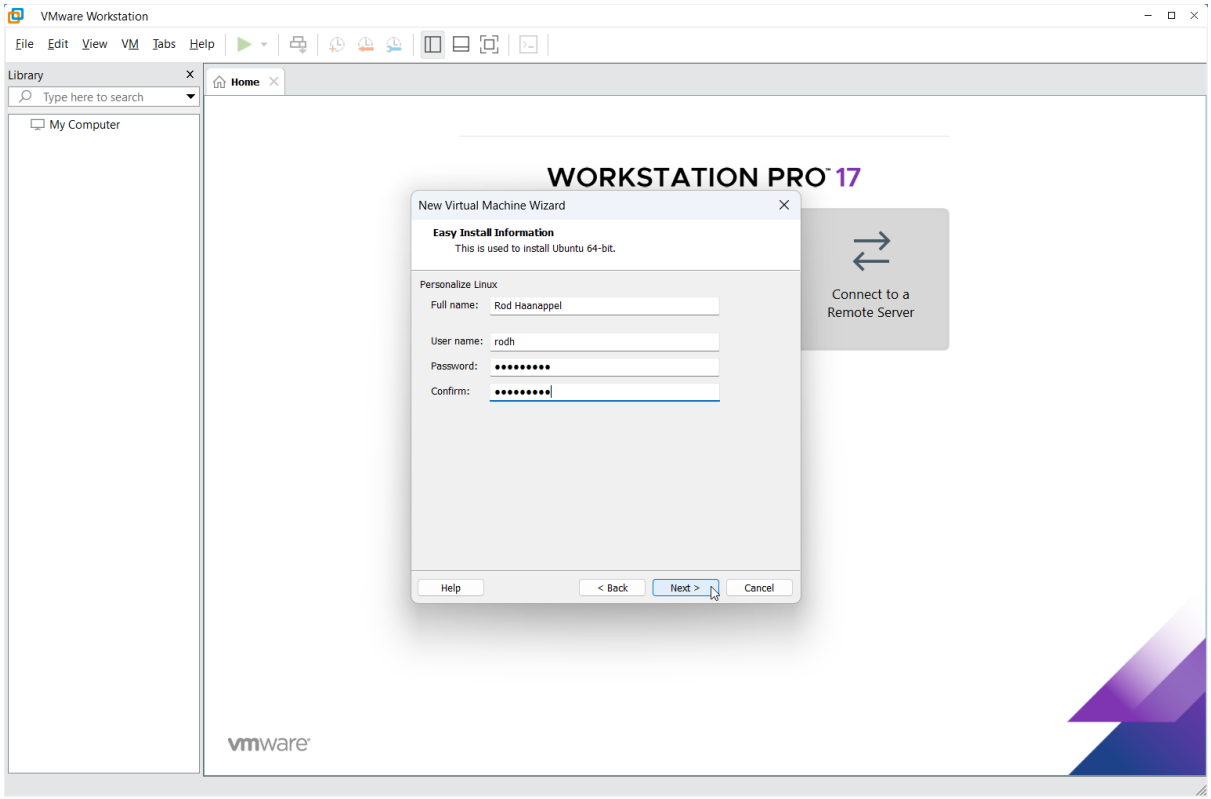



1.6. With the ISO file now displayed in the box and detected, click Next.

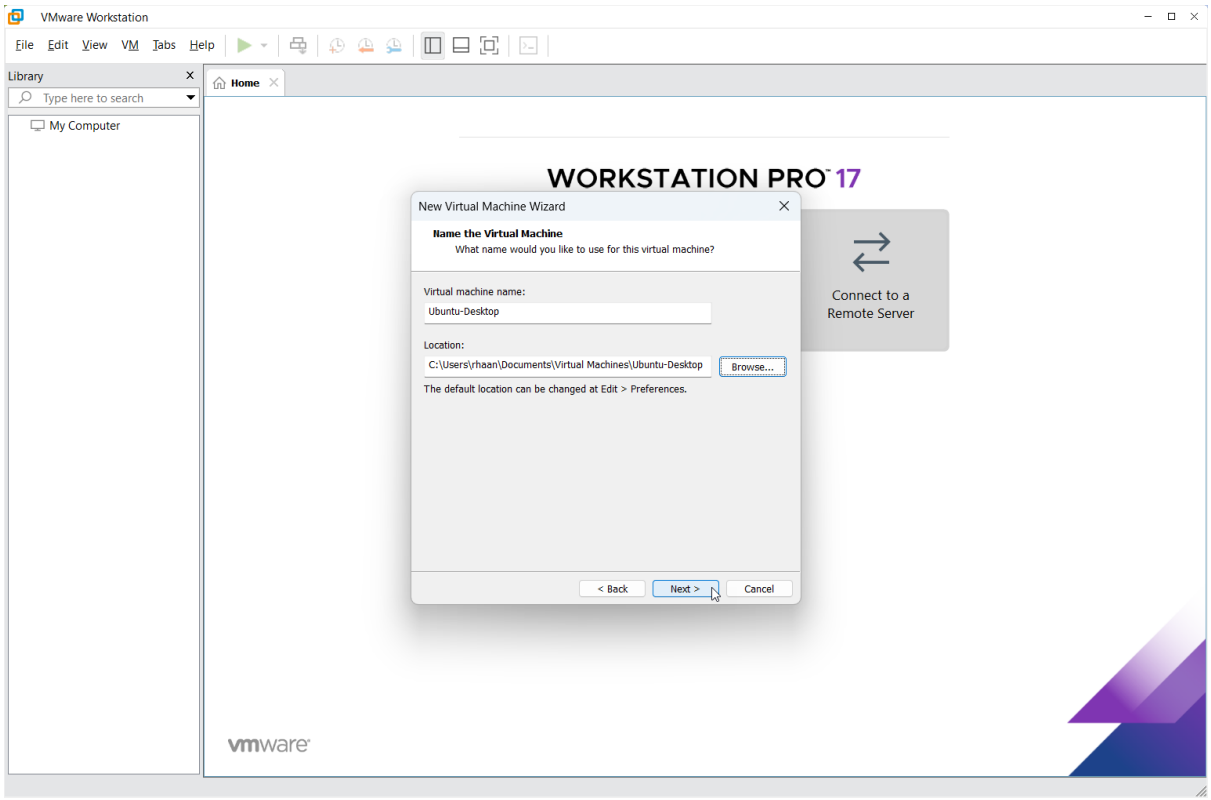



1.7. Create a personalization profile for Linux installs. To do this, fill out your full name, username, and password. REMEMBER the USERNAME and PASSWORD.

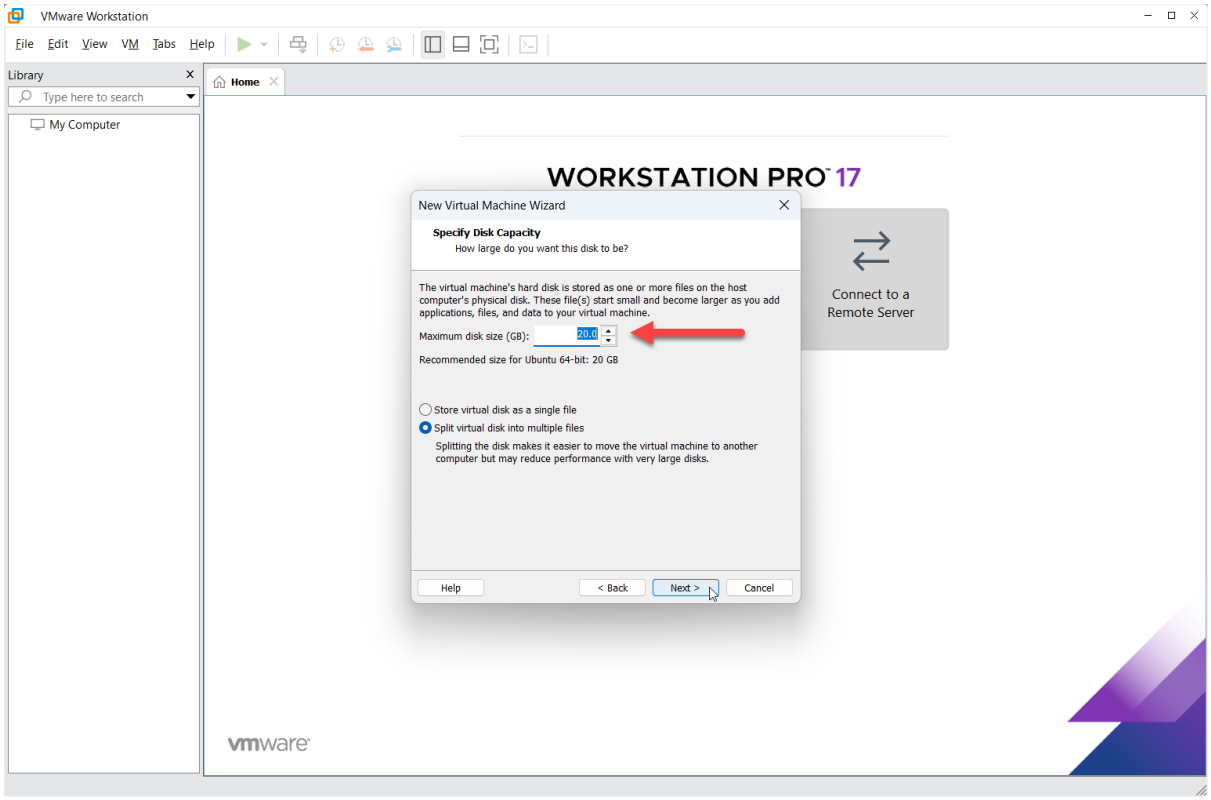
1.8. Click **Next**.



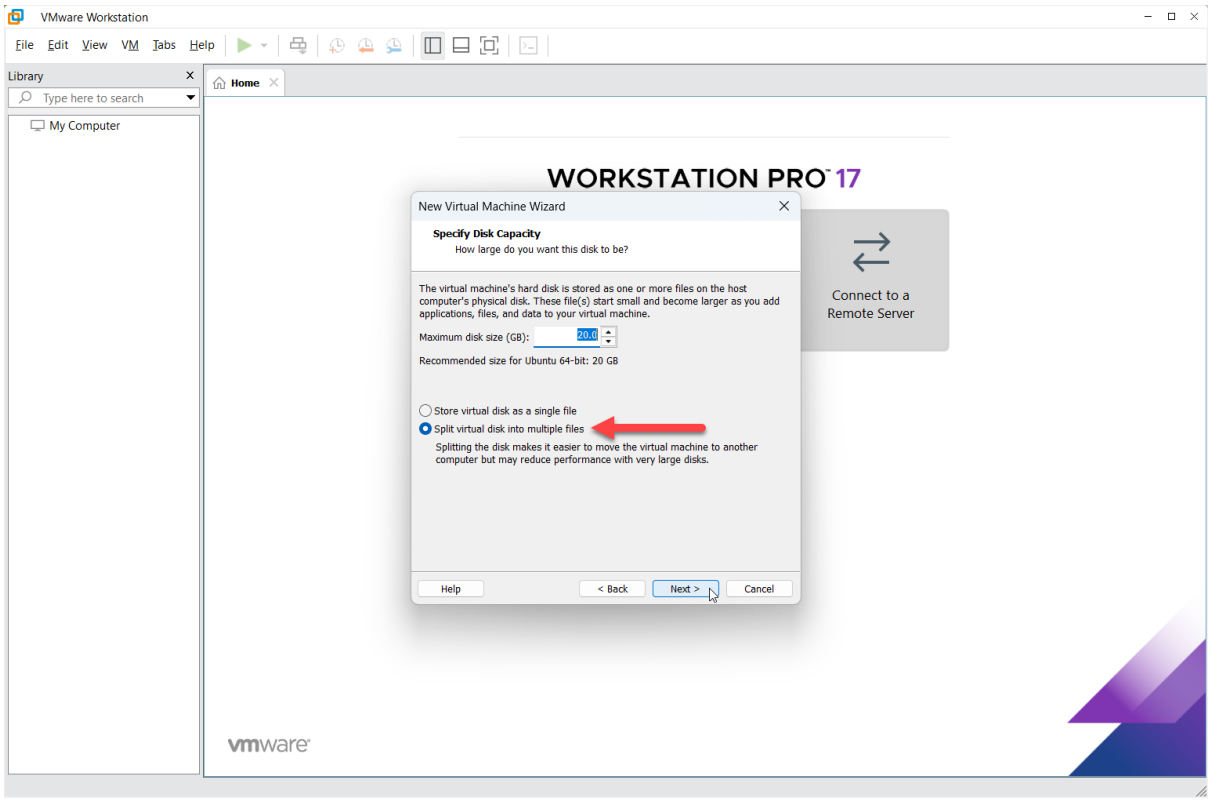

1.9. Enter a display name for your Ubuntu virtual machine. I recommend **Ubuntu-Desktop** (this name is for VMWare purposes only. The Ubuntu machine's hostname will be configured later). Click **Next**.



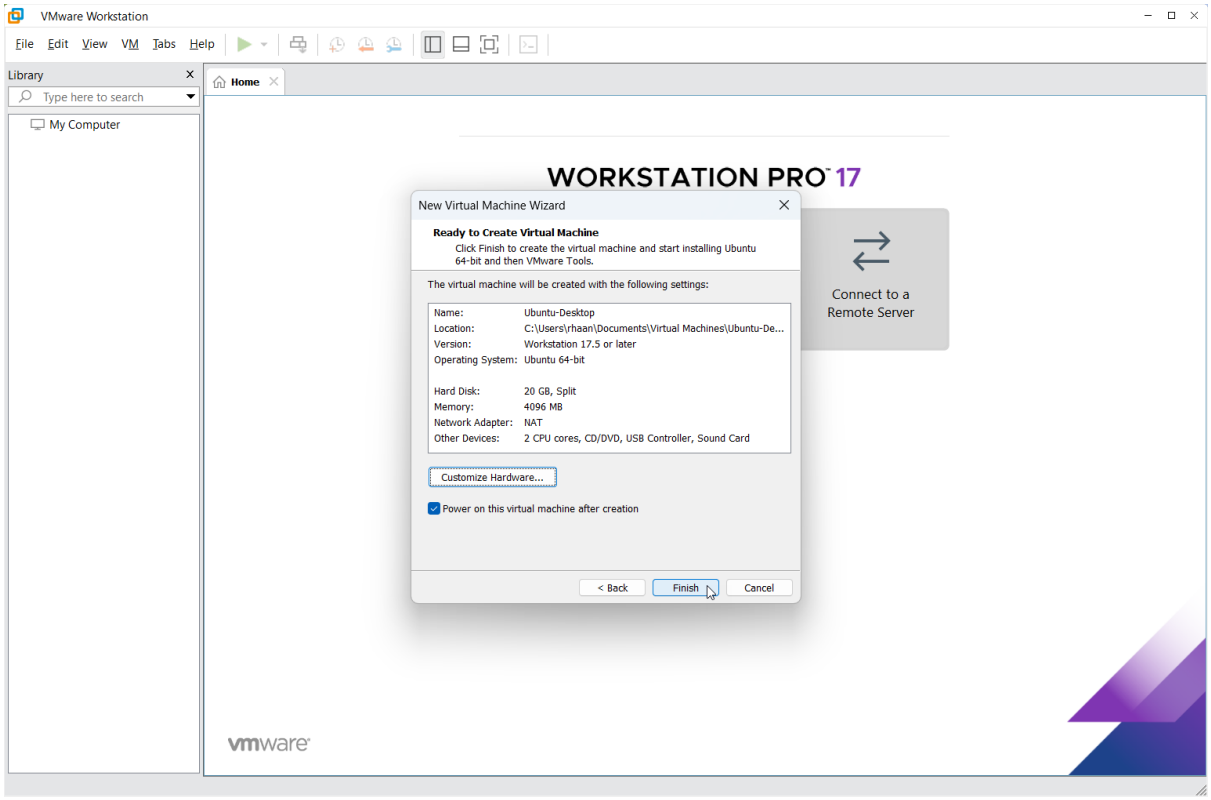
1.10. Specify a size for the virtual machine disk. The default is **20GB** if your personal machine (laptop or desktop) has sufficient free disk space. Leave at the default value because 20GB will not be blocked - only the actual amount of disk space used by the VM.



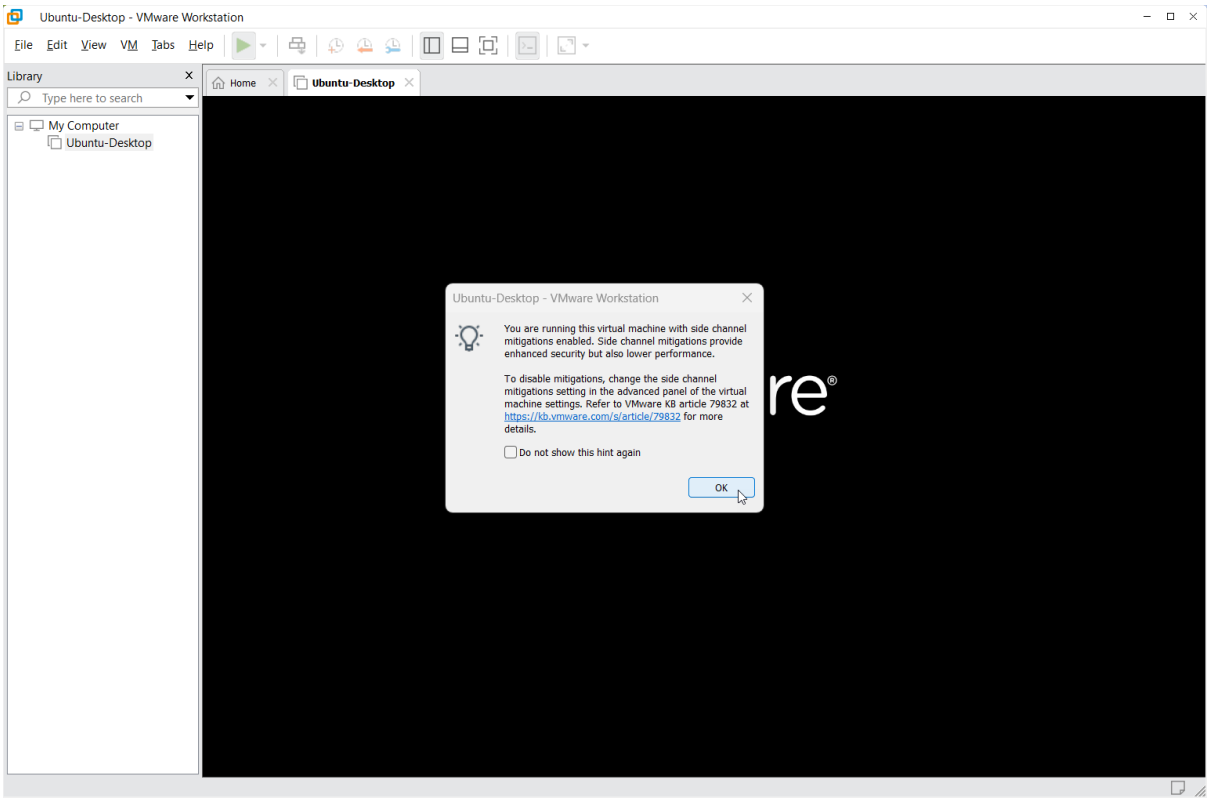
1.11. Select the option to **Split virtual disk into multiple files**. Click **Next**.



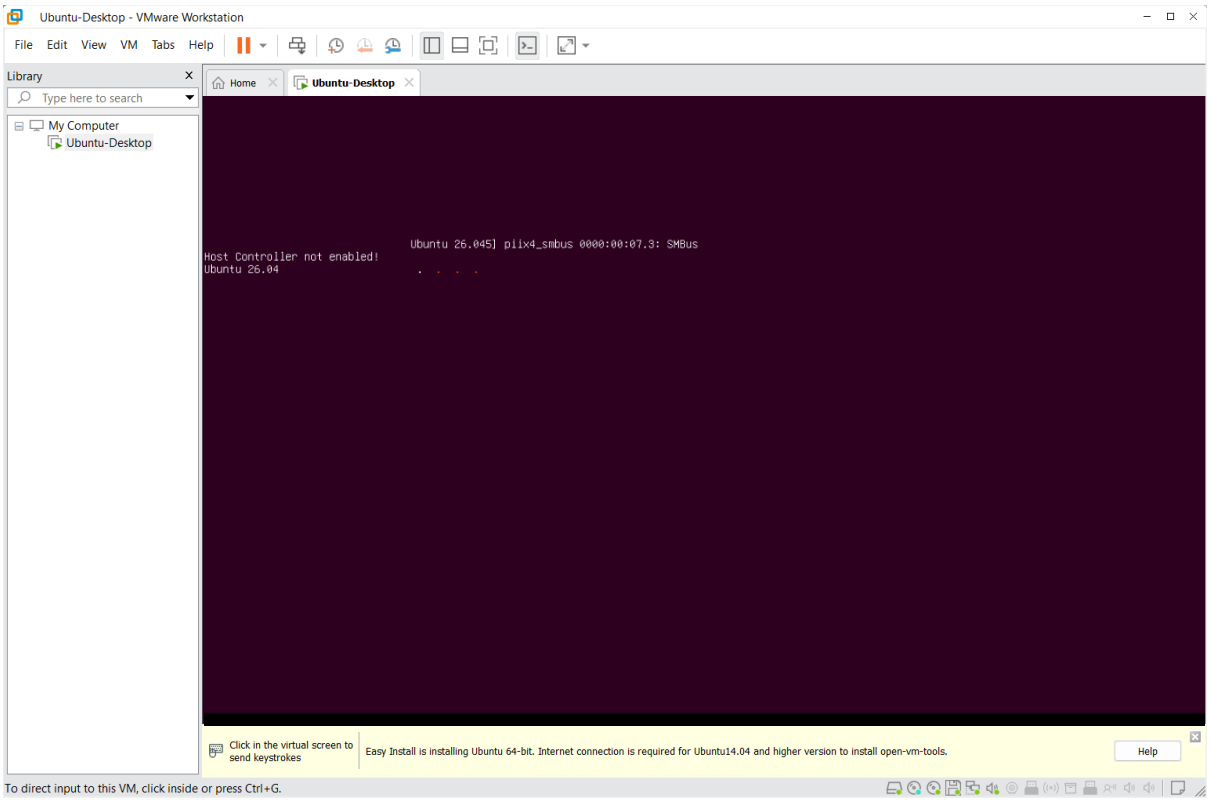

1.12. Select the option to **Power on this virtual machine after creation**. Click **Finish**.



1.13. The Ubuntu VM will now be created. You may get a pop-up message warning about "*side channel mitigations*". Ignore for now and click **OK**.



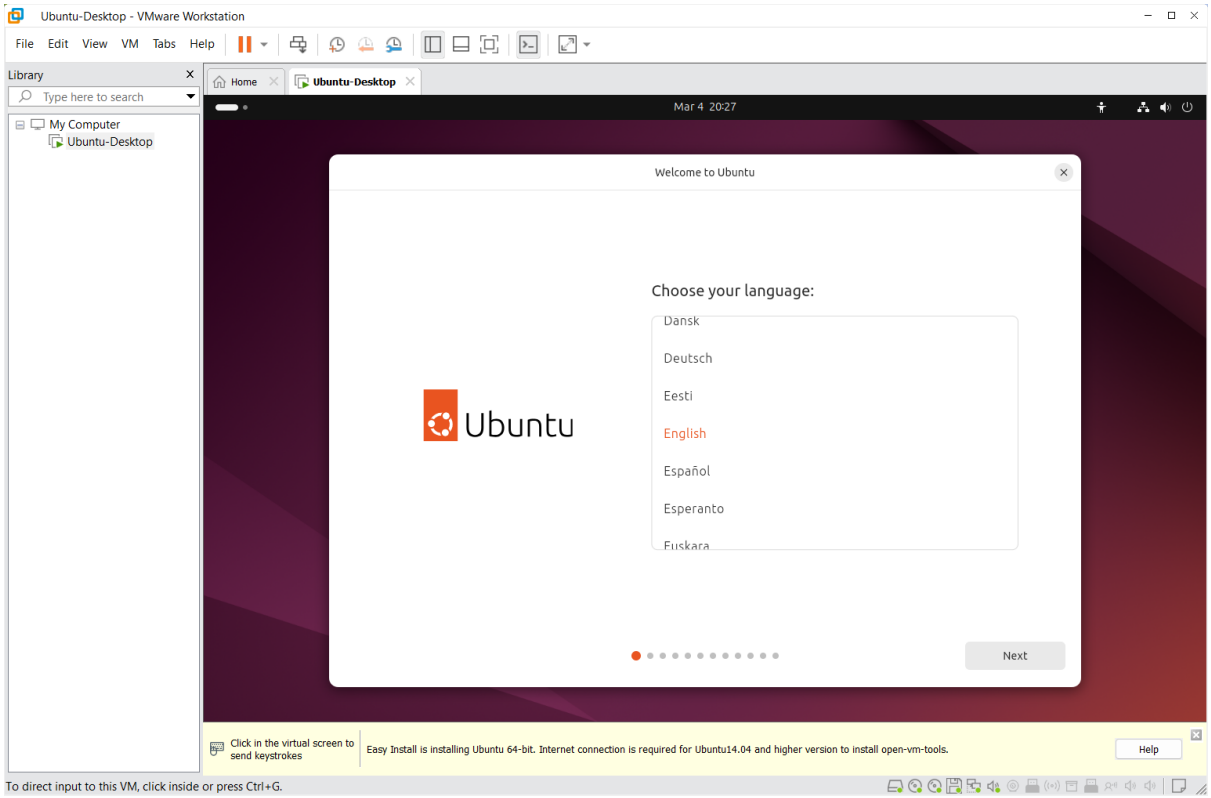

1.14. Allow the VM creation process to complete.



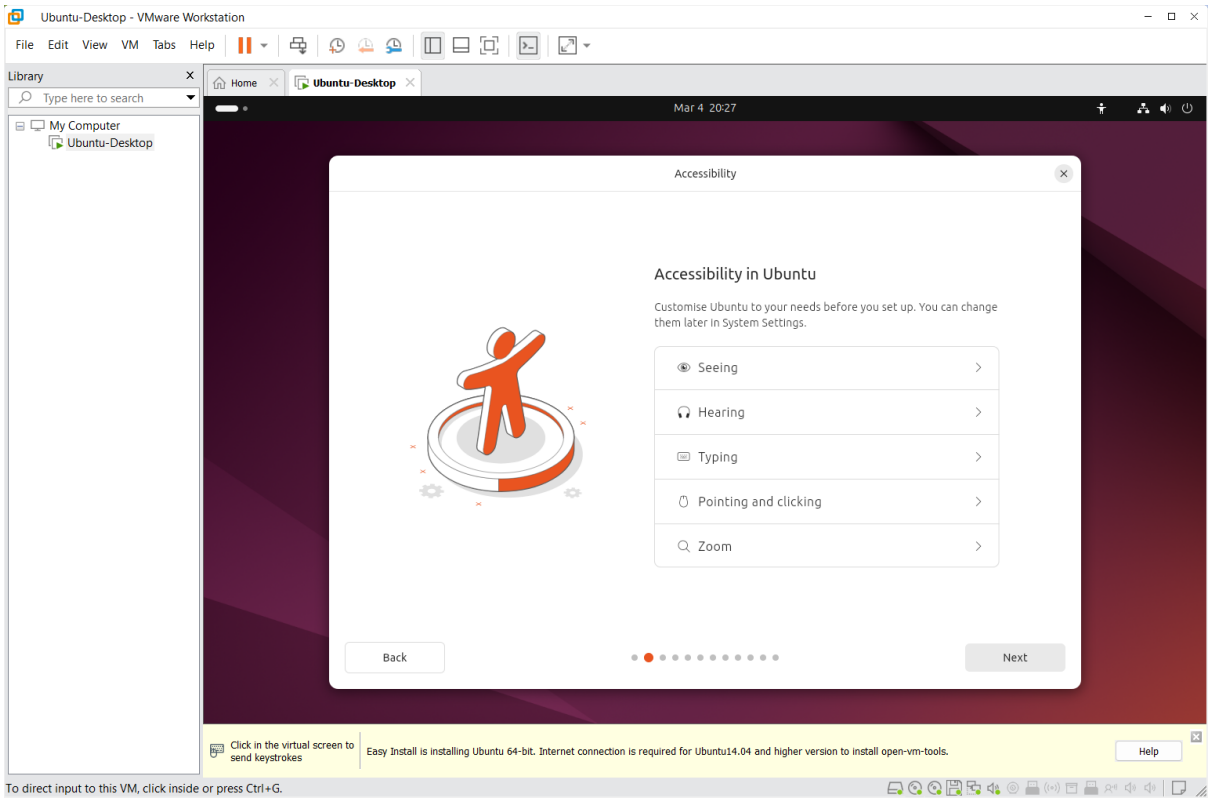
# 2. Setup and configure the virtual machine operating system

## 2.1. Setup and configure the virtual machine operating system (Ubuntu Desktop 26.04.1 LTS).

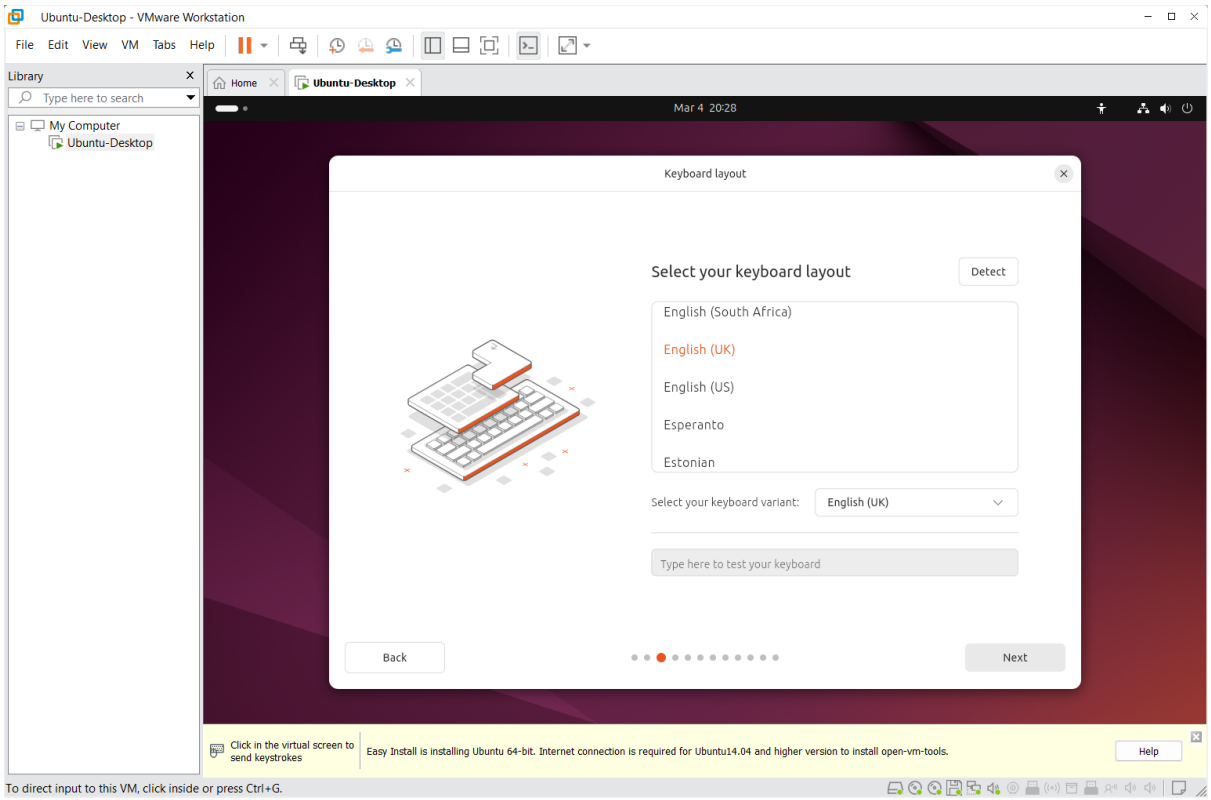
When VM creation has completed, you are required to progress through configuration prompts. The first is **Language**. Select your preferred language and click **Next**.



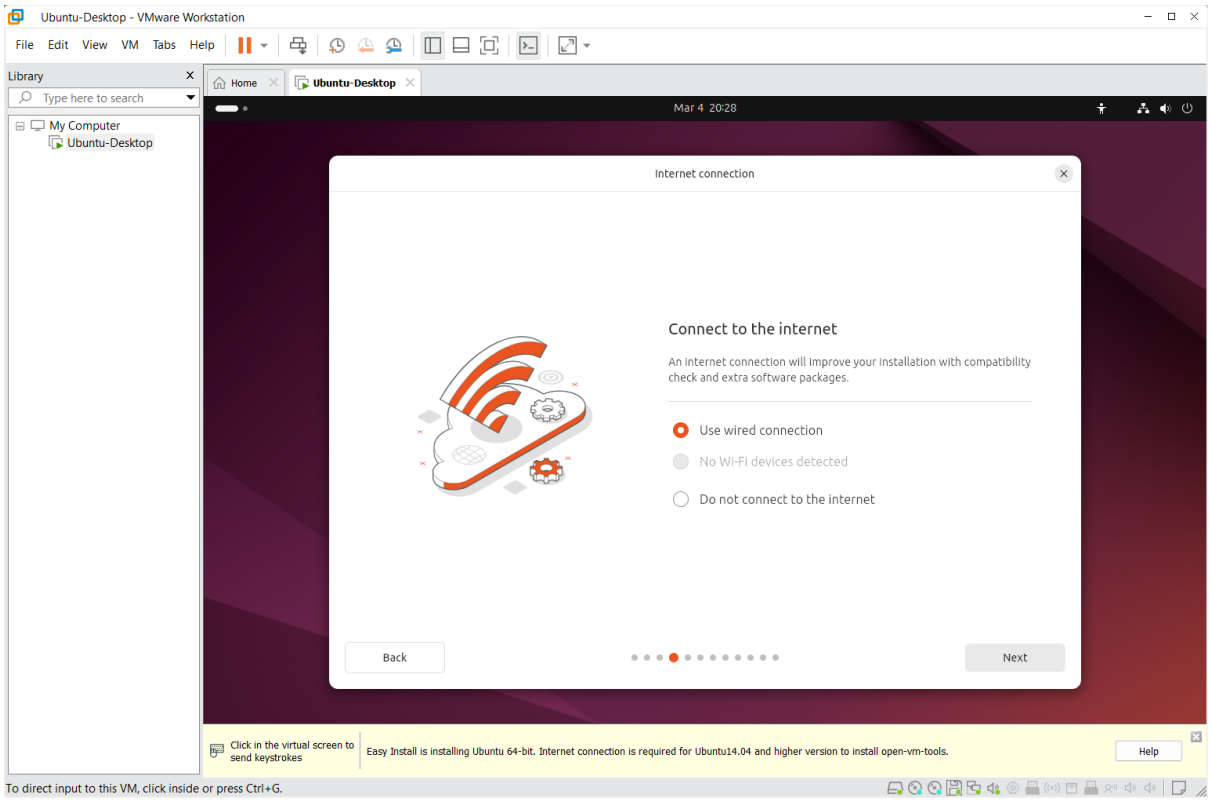
2.2. Unless you require **Accessibility** features, you can leave this page unconfigured and click **Next**.



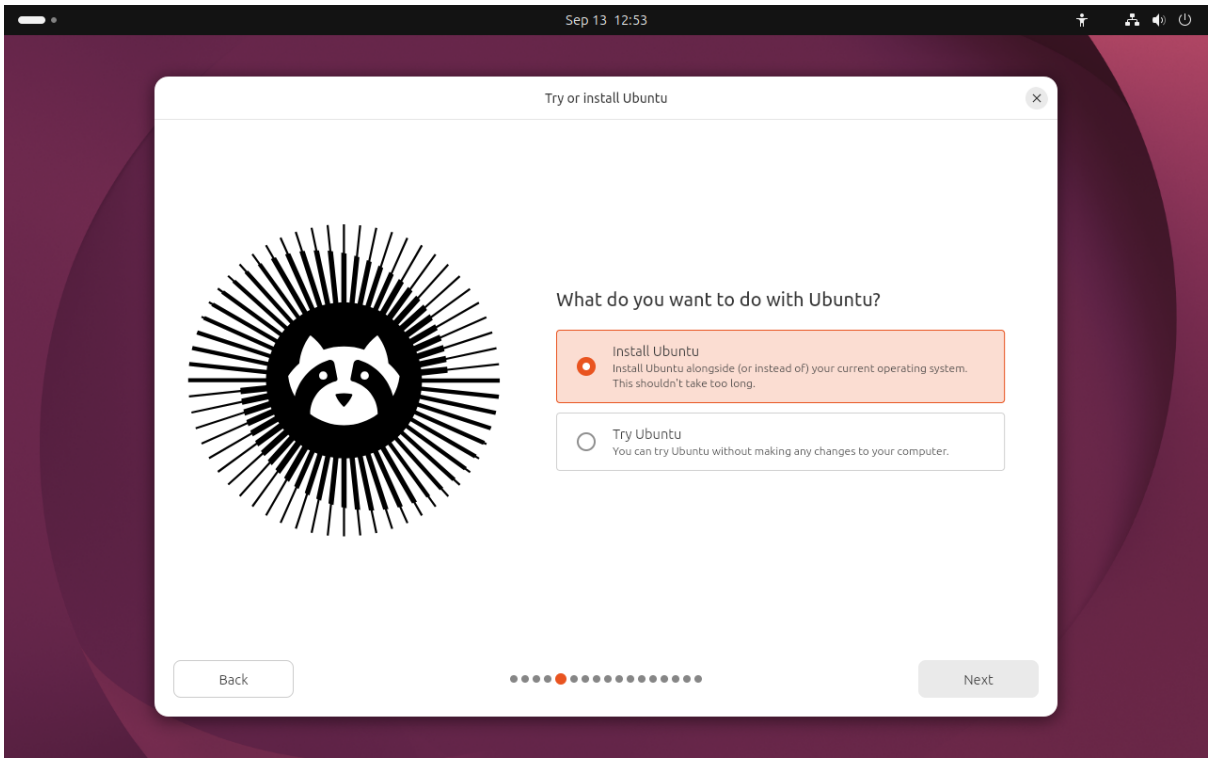

2.3. Select the **keyboard** layout you prefer to use on *your laptop*, and click **Next**.



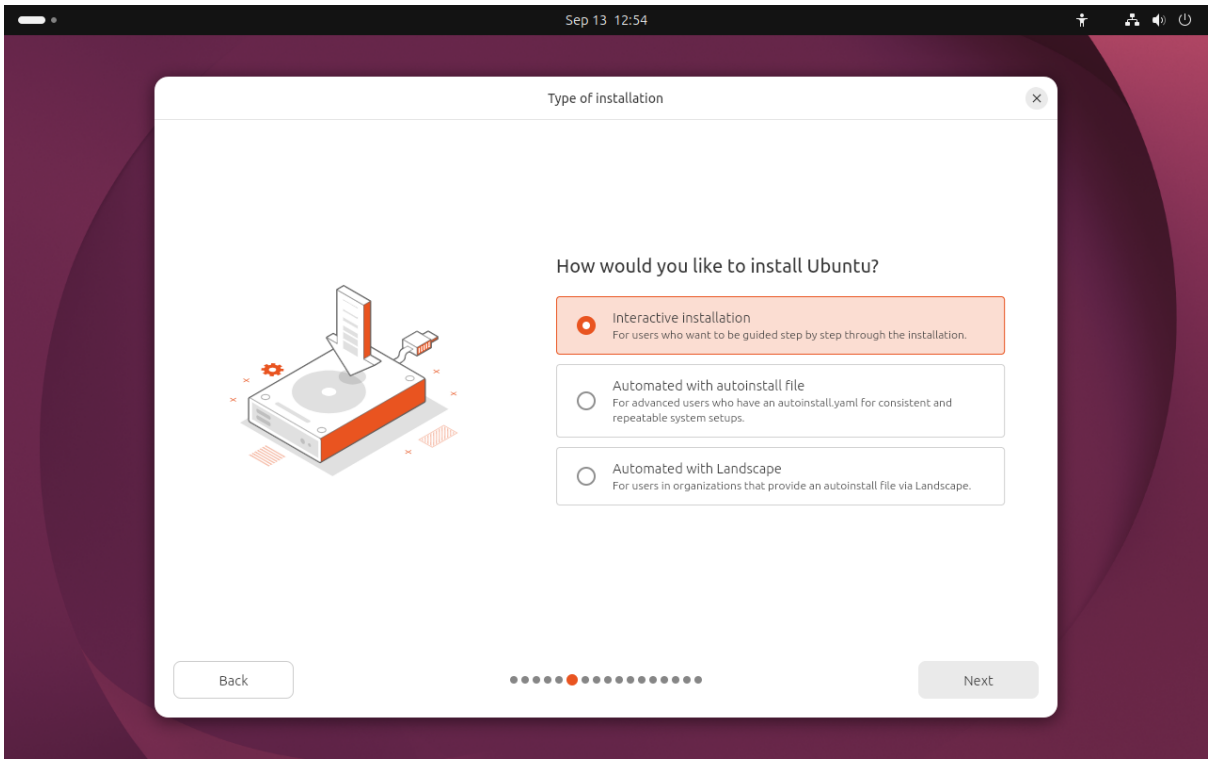
2.4. Even if the physical machine (laptop or desktop) on which VMWare is installed is connected to WiFi, the virtual machine is directly connected to the physical machine. Therefore, select the **Use wired connection** option. Click **Next**.



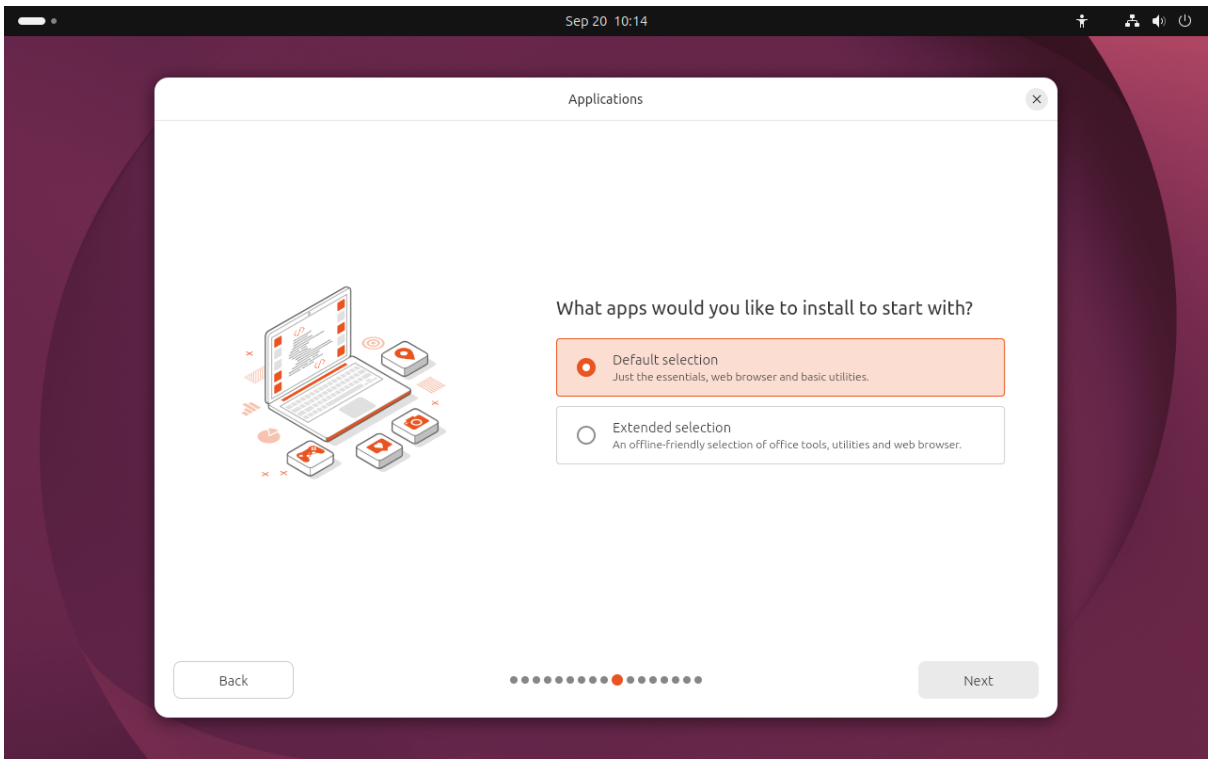

2.5. Select **Install Ubuntu** and click **Next**.



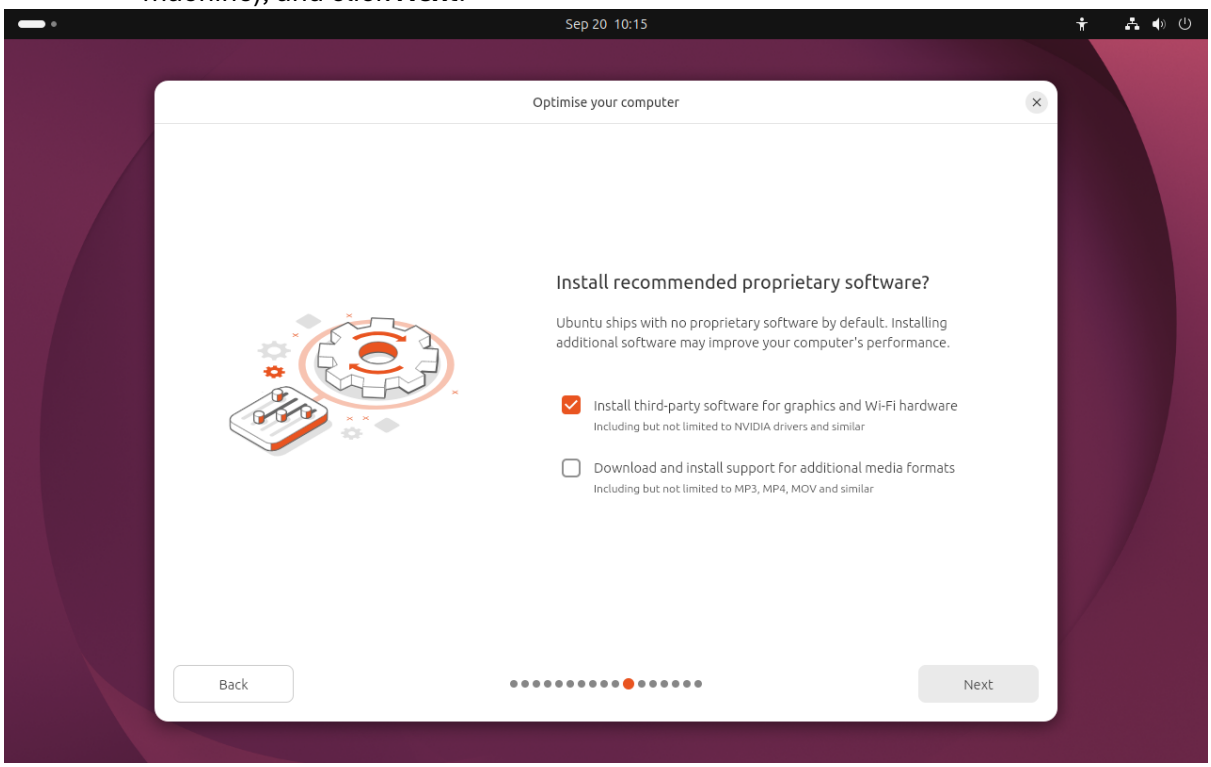
2.6. Select **Interactive installation** and click **Next.**



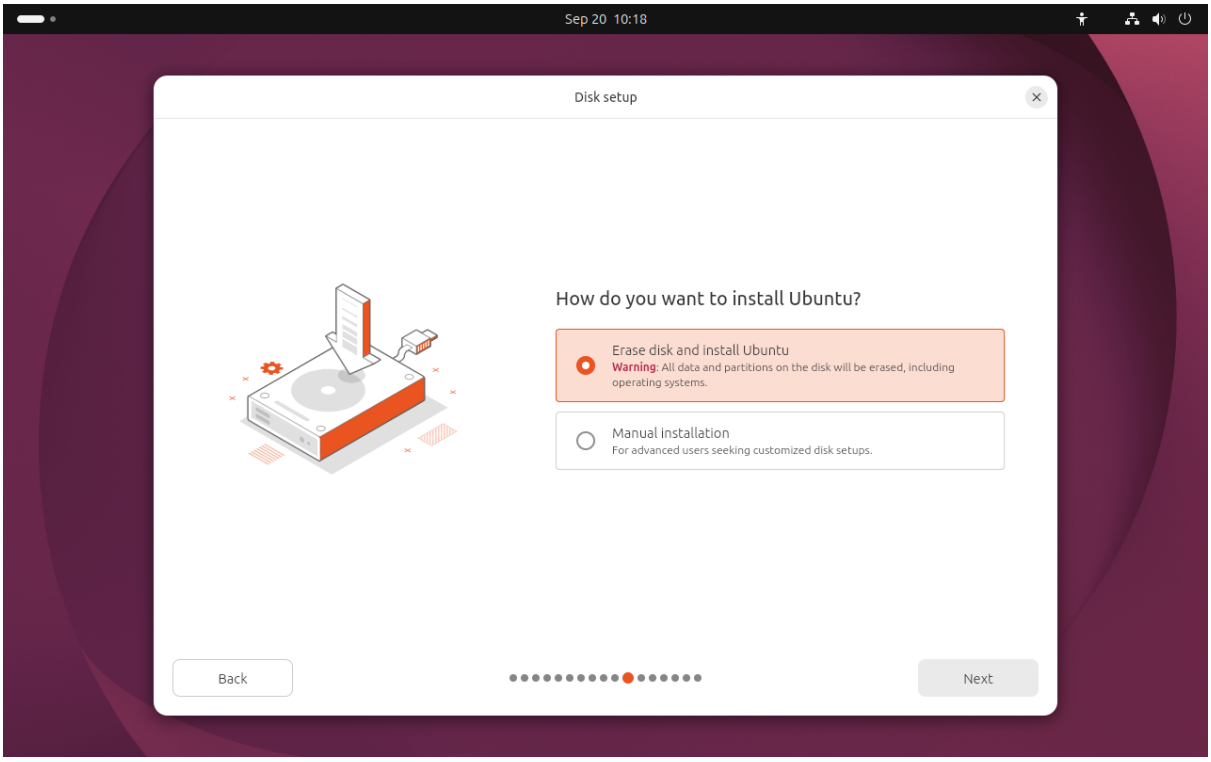

2.7. Choose **Default selection** and click **Next.**



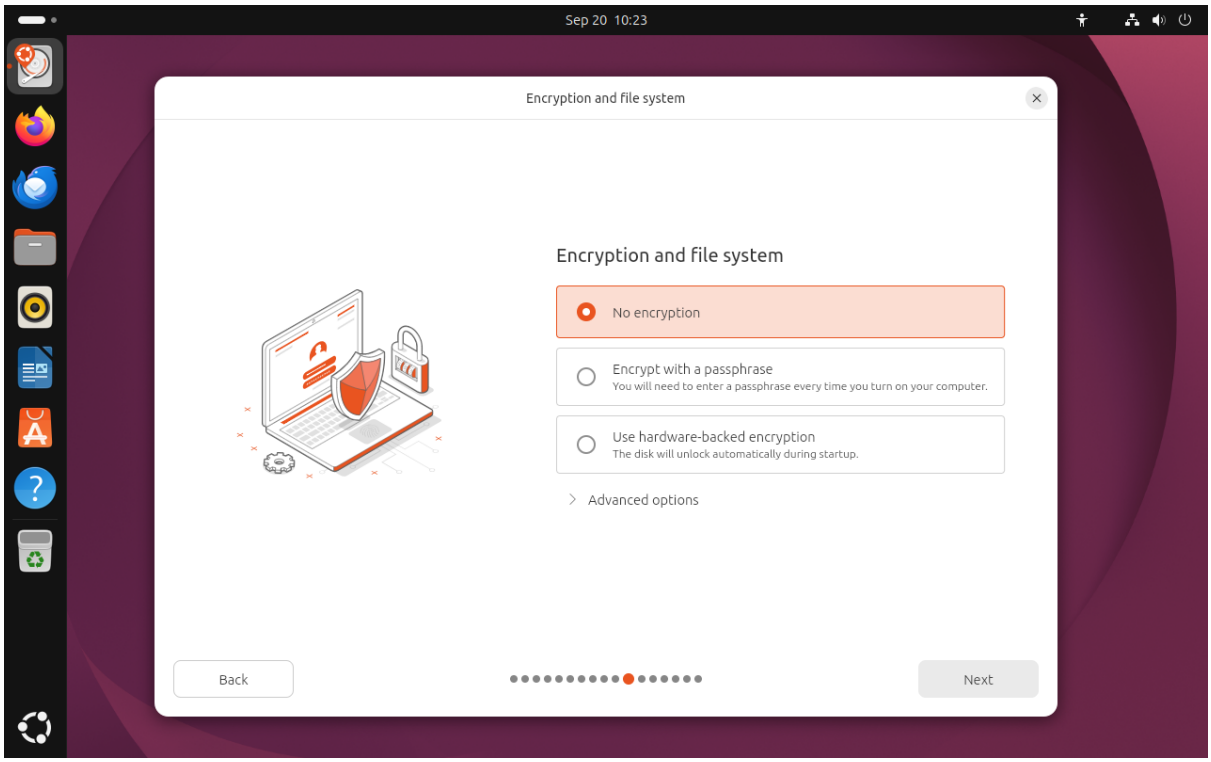
2.8. Choose to **Install third-party software** (to support hardware specfic to your host machine), and click **Next**.




2.9. Select **Erase disk and install Ubuntu**. Don't panic, this refers to the VM disk, not your host machine (i.e. laptop disk).

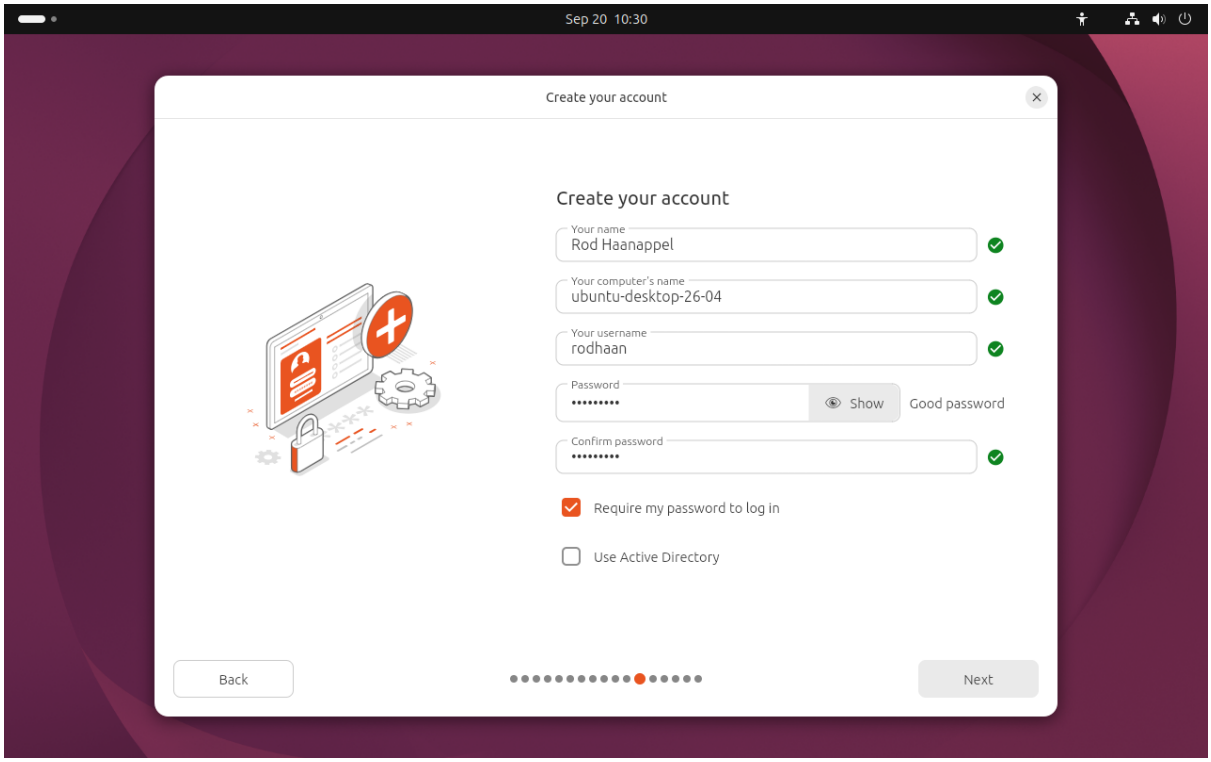


2.10. Good security practice is to encrypt the data on your machine. However, in this case we will keep it simple and choose **No encryption**. Click **Next**.

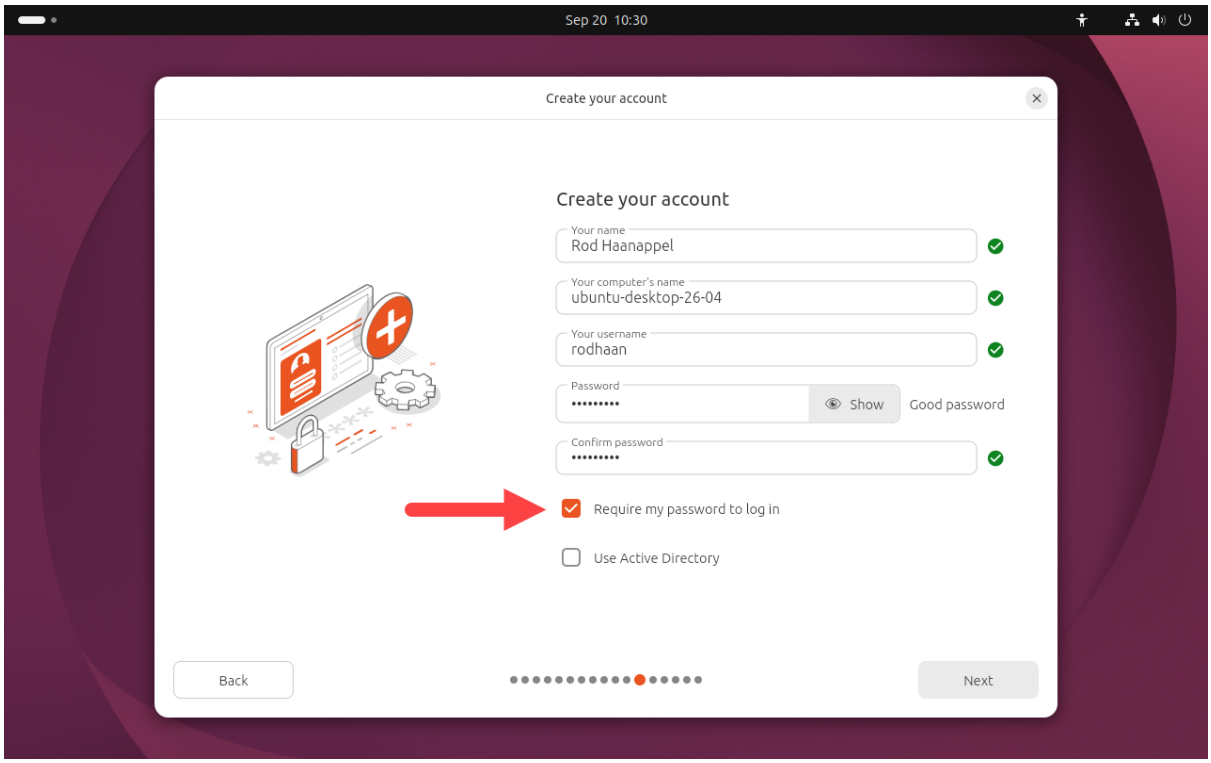


2.11. On the **Create your account** screen, provide the details for this machine. Change the "*Your computer's name*" to **Ubuntu-Desktop-26-04**.

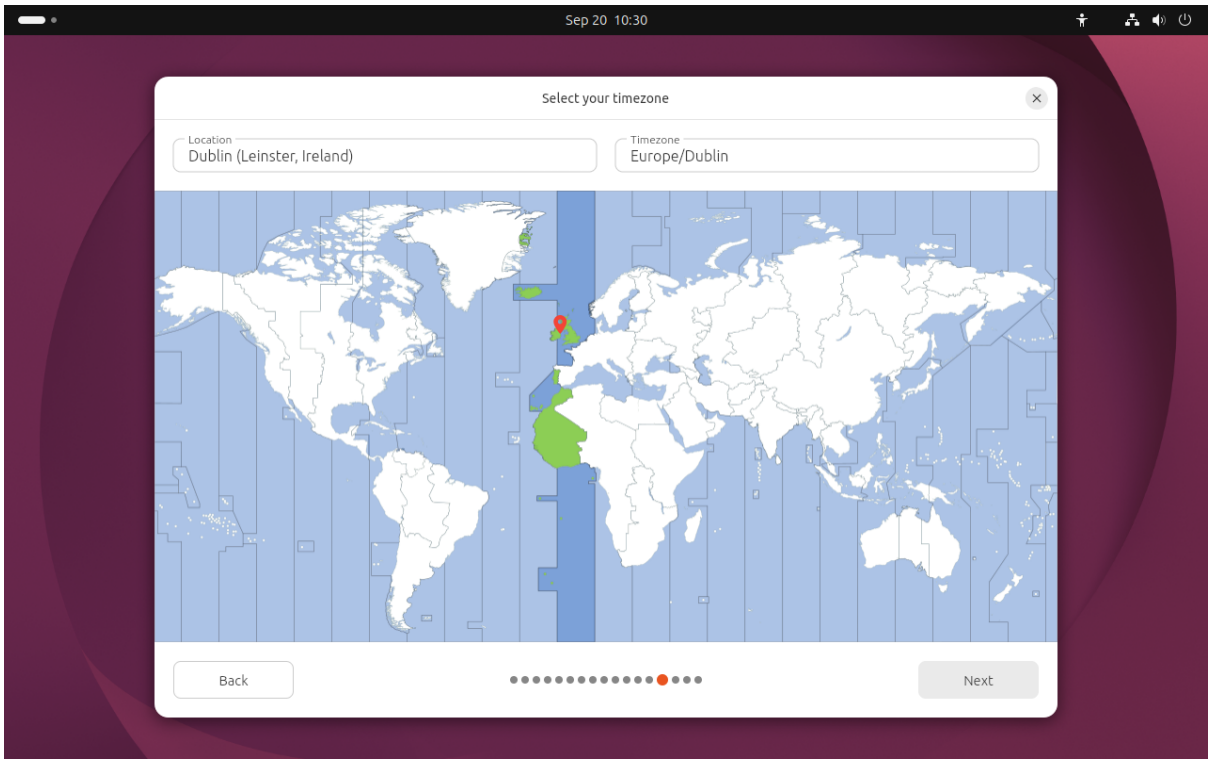
Create a **username** and **password**. Be sure to remember these as you will need to know your logon credentials when logging onto the Ubuntu VM.



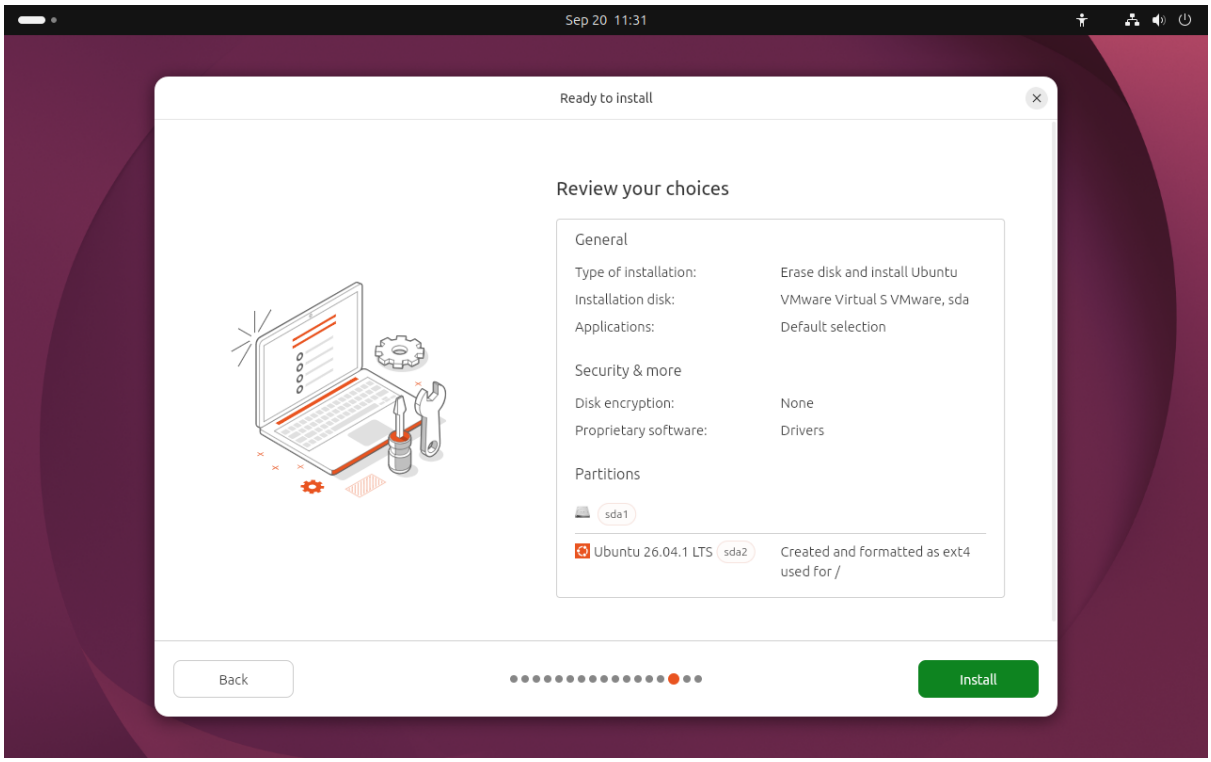
2.12. Enable the **Require my password to log in** option and click **Next**.

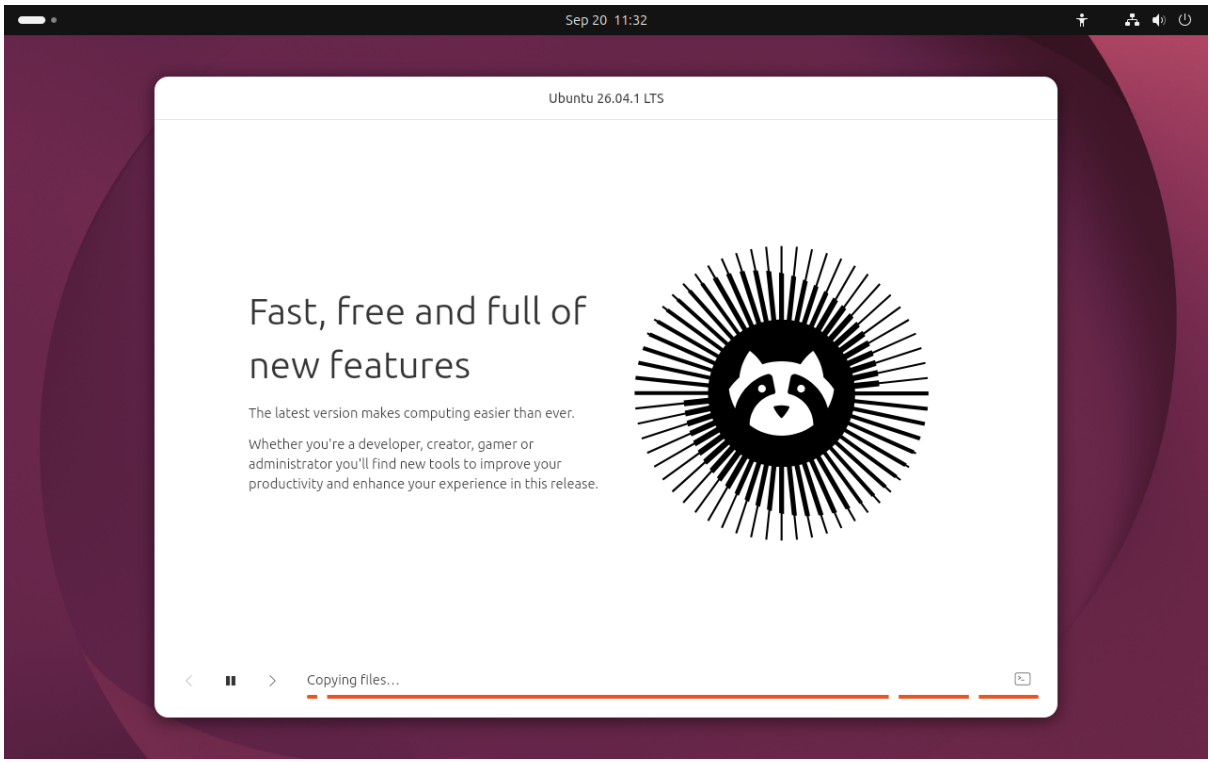



2.13. Select the time zone you are in (should be automatically detected) and click **Next**.

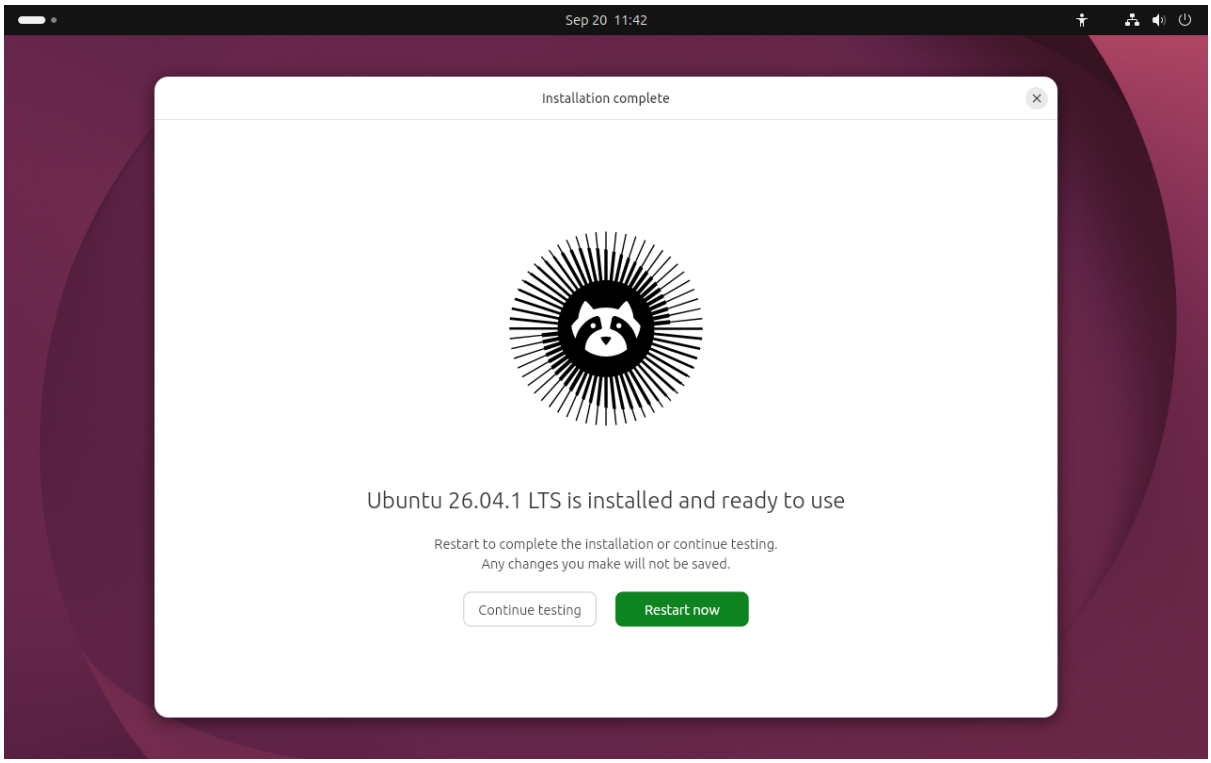


2.14. Review your choices and click **Install**. Ubuntu Desktop will now progress through the installation.

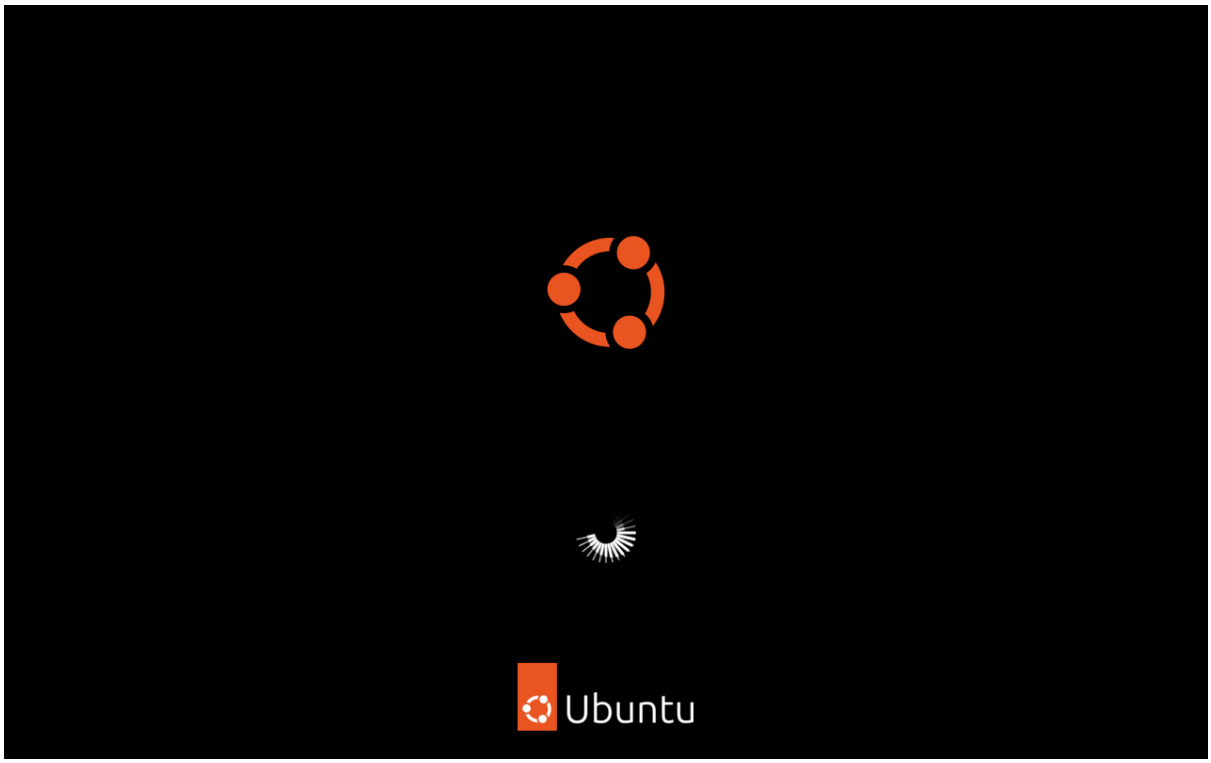


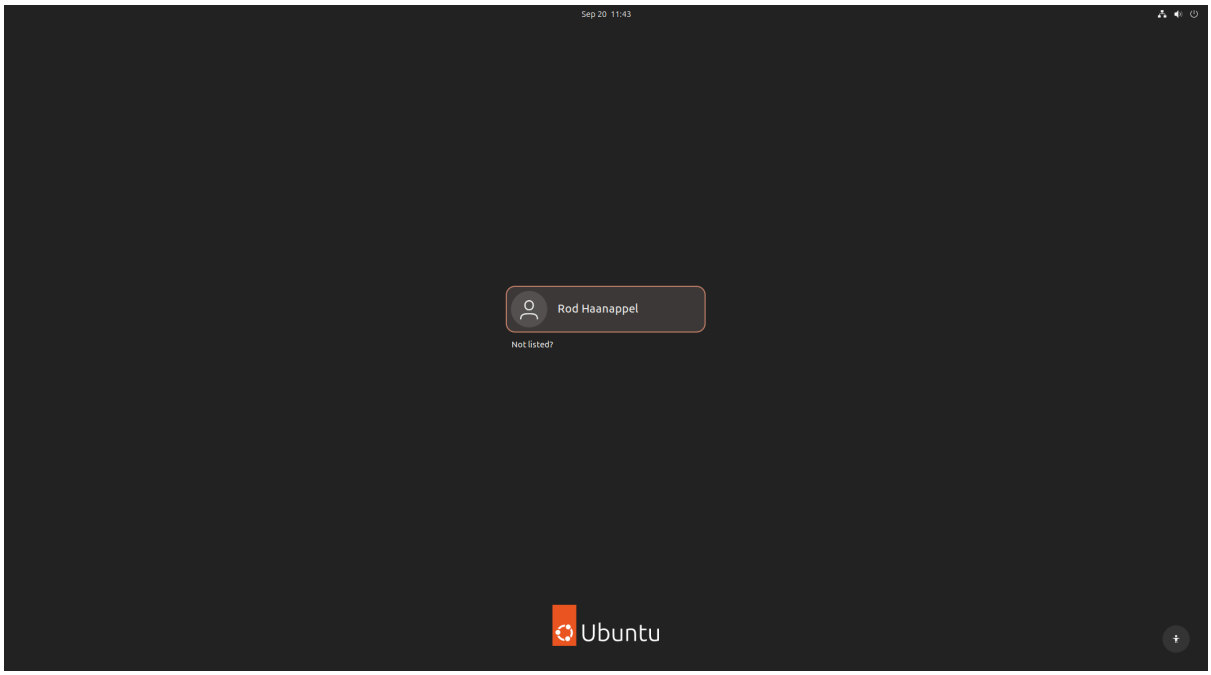
2.15. When the installation is complete, click the option to **Restart now**.



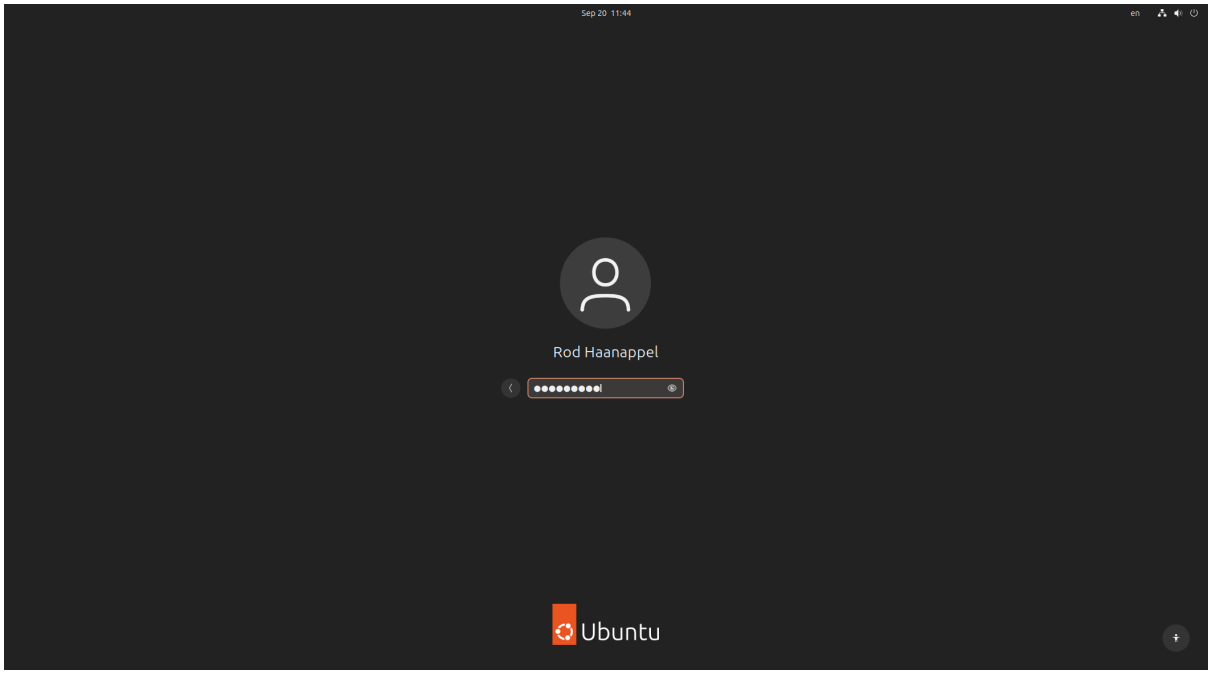

2.16. The VM will progress through the restart process.



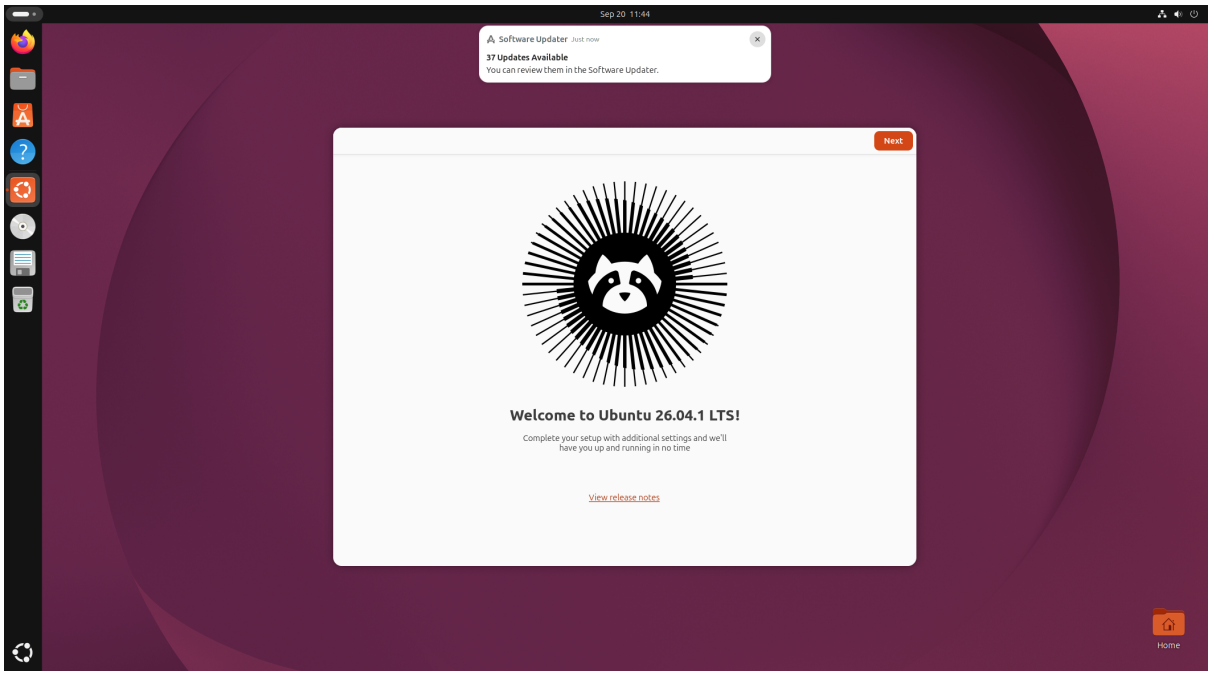
2.17. Click on your username.



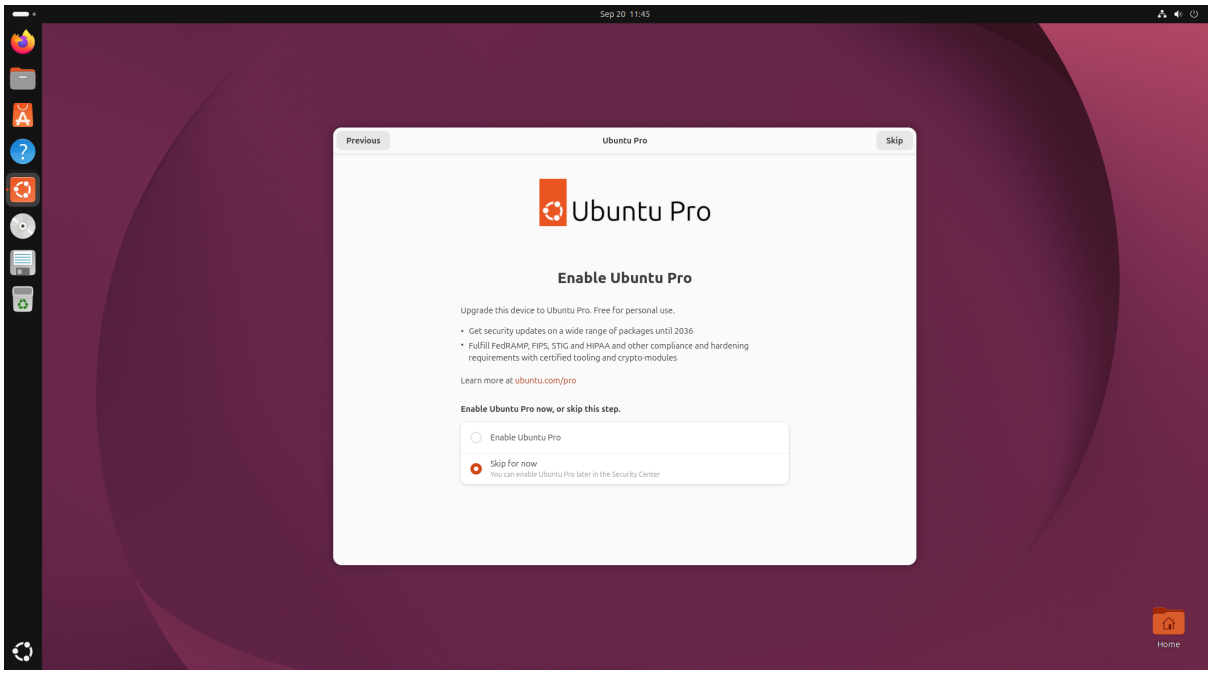

2.18. Enter your password and press the **Enter** key.



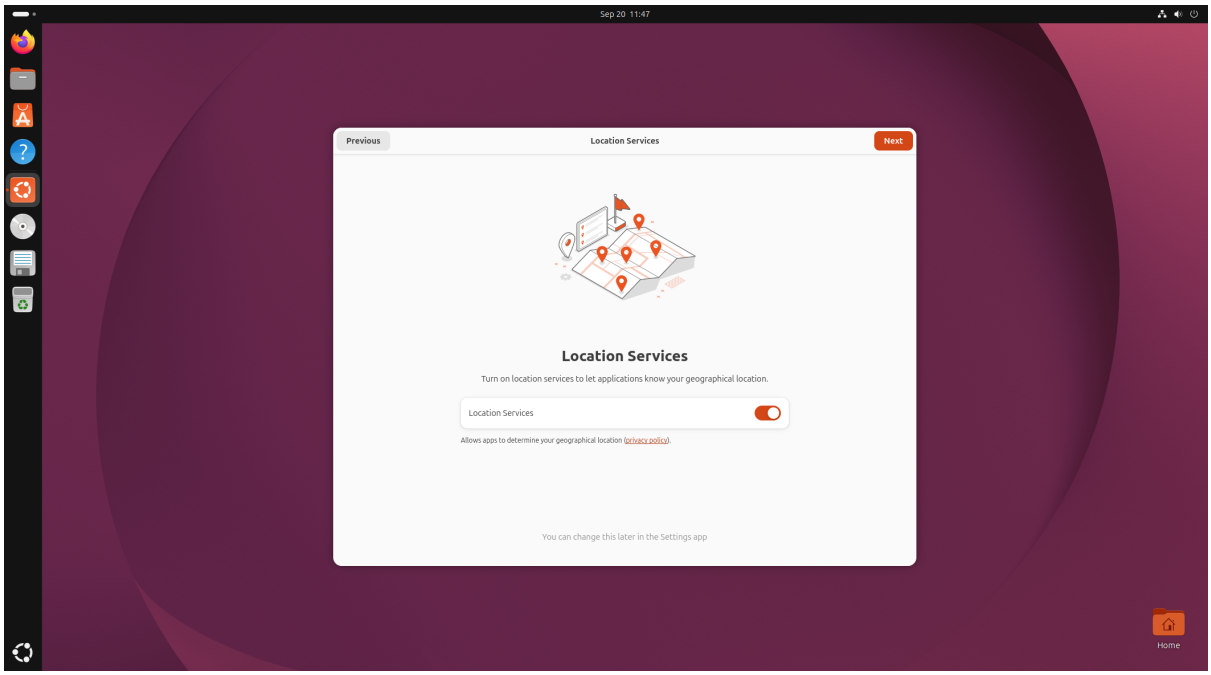
2.19. On the desktop, click **Next** on the **Welcome** screen.



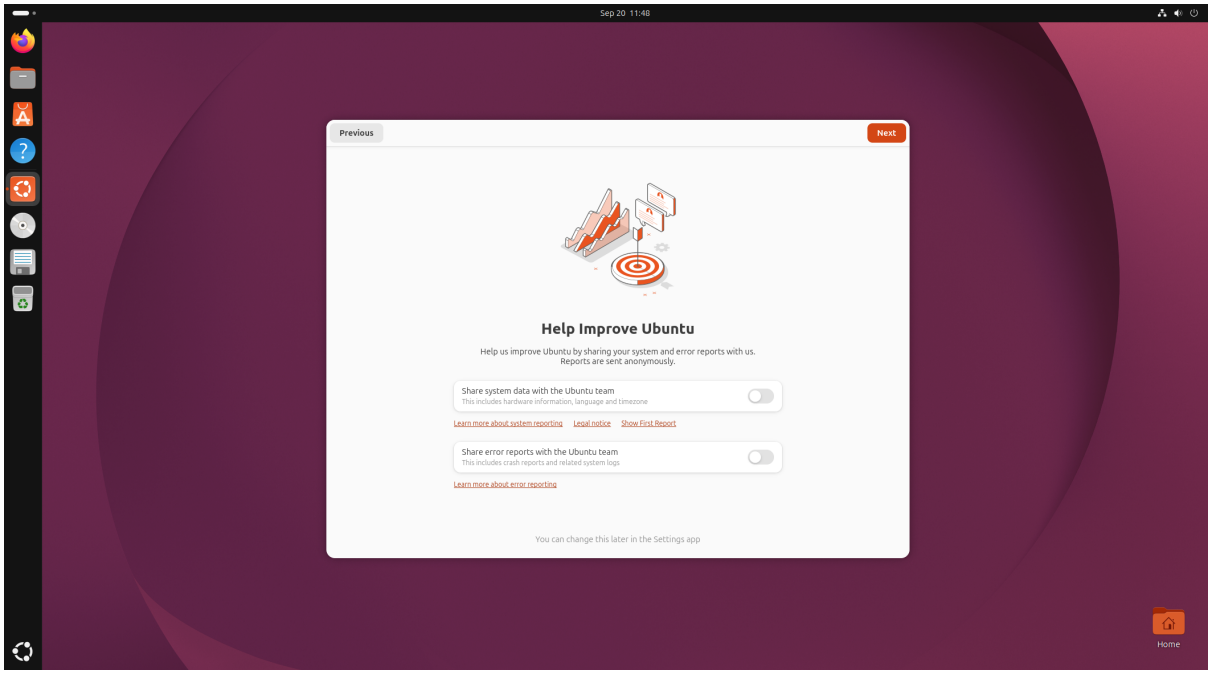

2.20. On the Ubuntu Pro screen, select **Skip for now** and click **Skip** at the top-right of the screen.



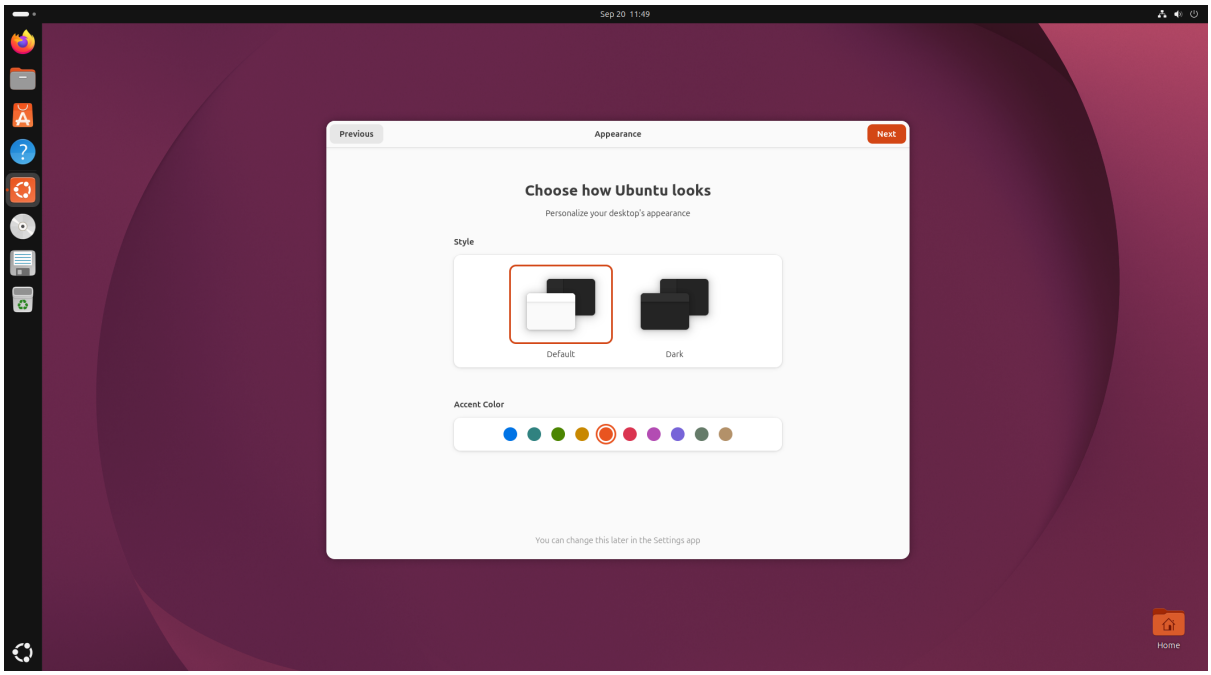
2.21. On the Location Services screen, enable **Location Services** and click Next.



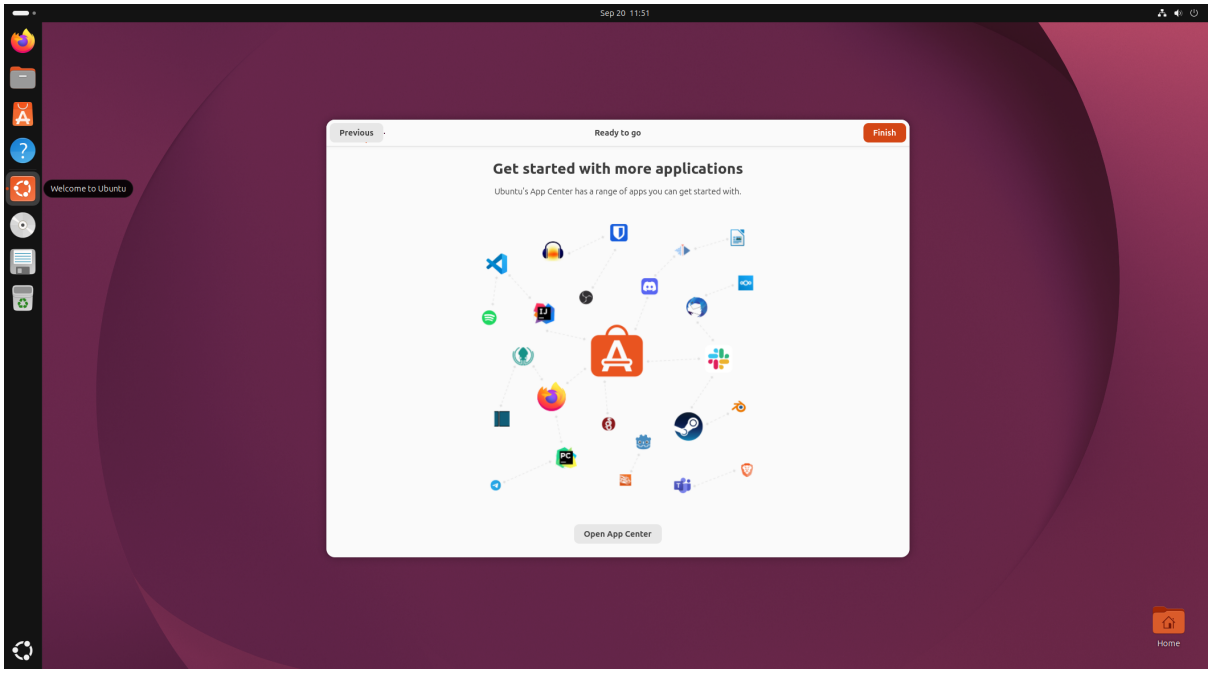

2.22. Select your preference for sharing system data with Ubuntu and click **Next**.



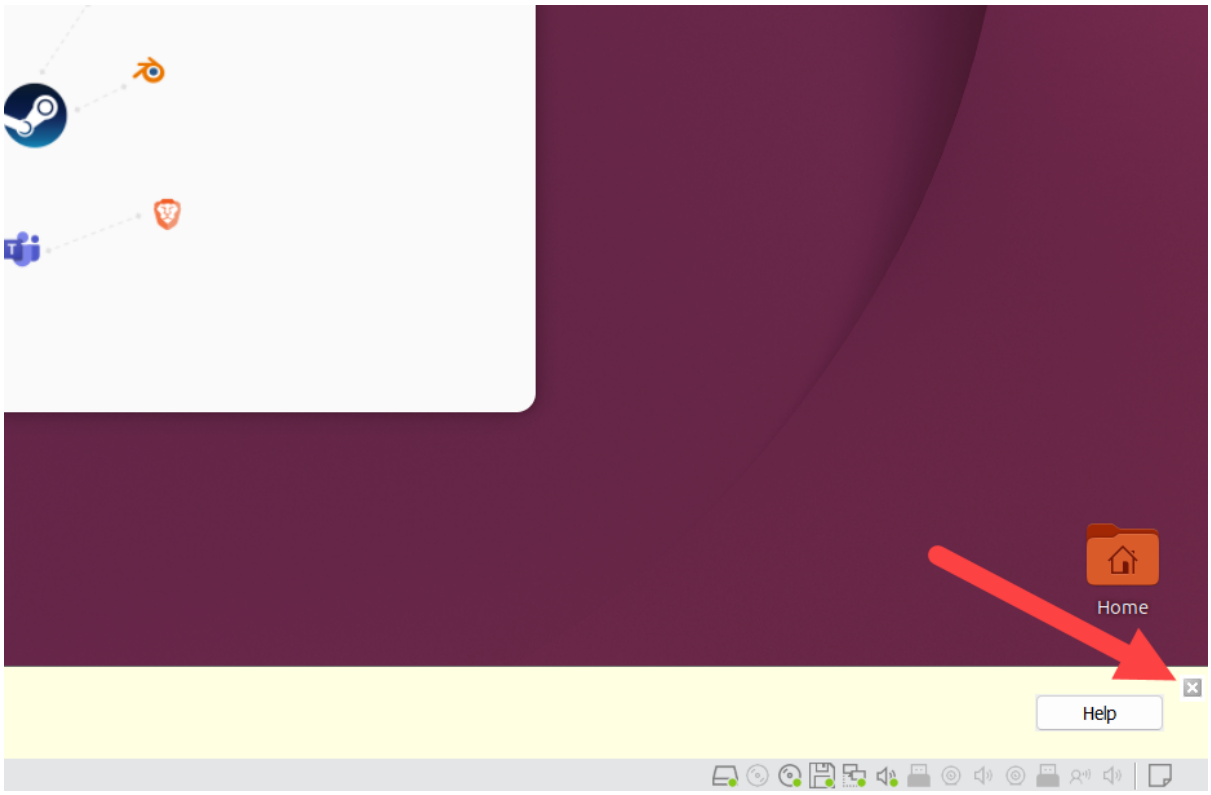
2.23. On the **Appearance** screen, keep the **Default** settings and click **Next**.




2.24. On the **Ready to go** screen, click **Finish**.



2.25. Close the "*Easy Install*" banner at the bottom of the screen by clicking the **X**.

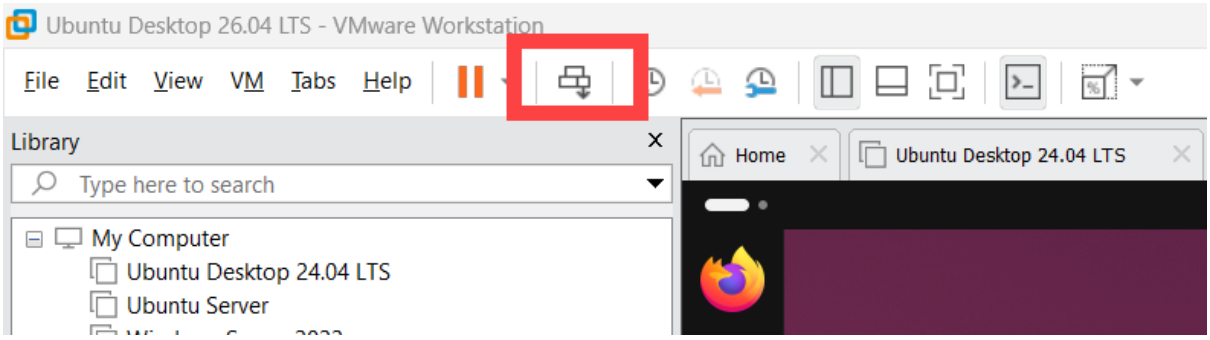



2.26. The Ubuntu Desktop installation and operating system configuration is now complete. Effectively we now have a customised Ubuntu Desktop machine.

## 3. Mouse Control and Power Control of your VM in VMware

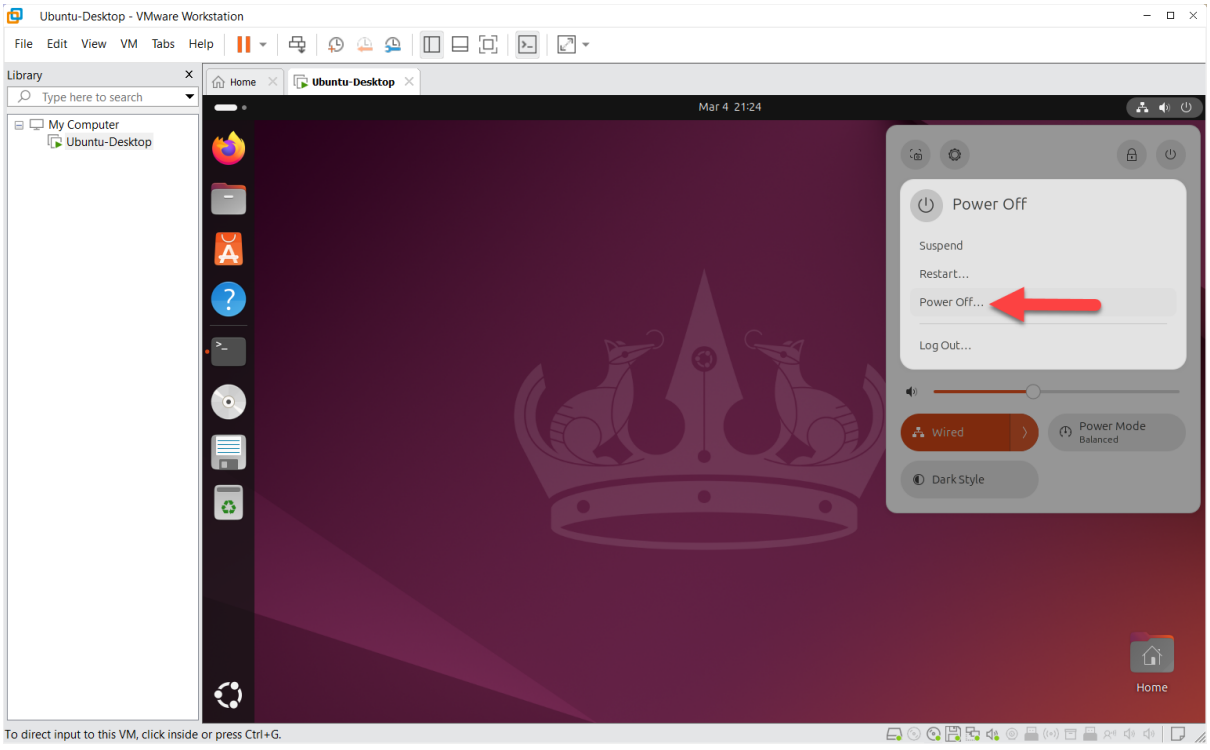
- On occasion, you may *lose* your mouse cursor. It gets stuck in the VM and you can't find it for use in your actual laptop's operating system. If this happens, just press the `Ctrl+Alt` keys (i.e. hold down the Ctrl key then press the Alt key).

- In some VMs, especially Windows OS, your need to press `Crtl+At+Del` keys to logon. However, this will be interpreted by your laptop's OS and will invoke the lock screen. To get around this problem VMware Workstation has a `Crtl+Alt+Del` shortcut button. So click this button when your VM requires it:

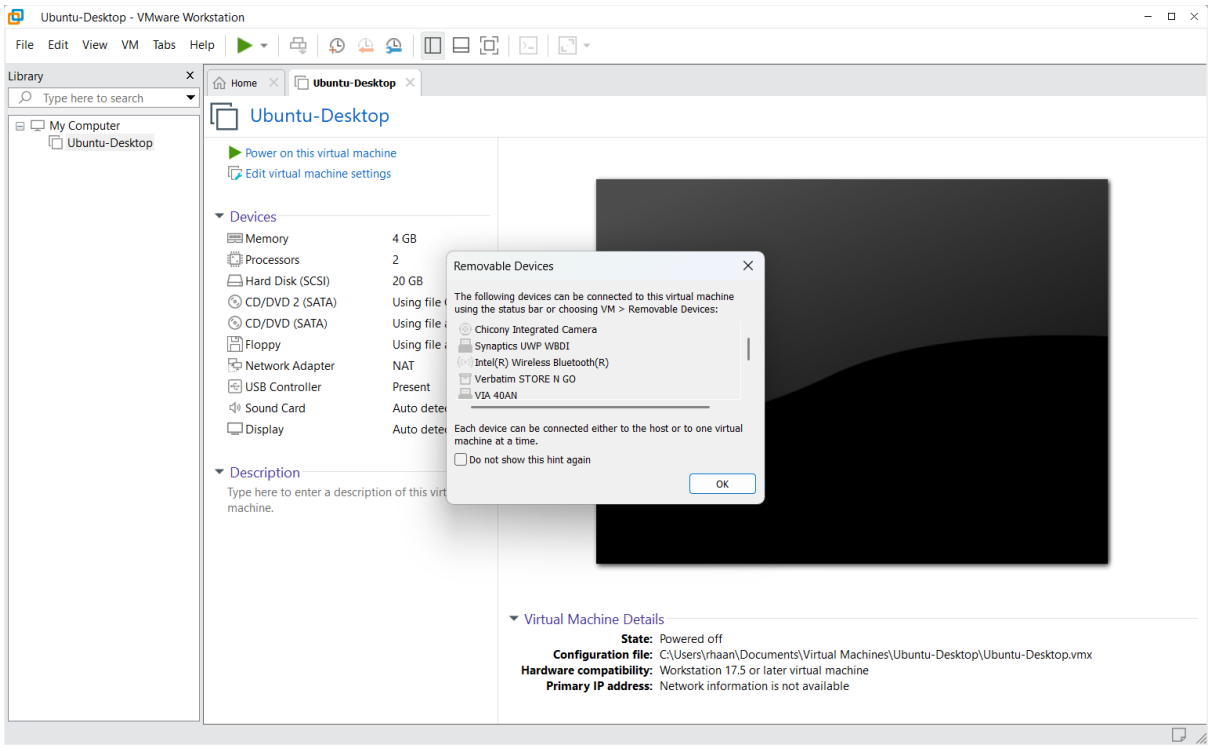



- Power control of your VM:

  Click the Power icon  at the top-right corner of the Ubuntu desktop, and select the **Power Off** option.



3.1. The VM is now powered off. If a window displays informing you that you can add physical devices as virtual devices to the VM, click **OK** to close the window.




3.2. You can exit **VMware Workstation** now (or **Fusion** if you're on a macOS system). Before closing VMware Workstation, it's always best practice to shut down all VMs first.

3.3. **Power on your VM**: There are several ways to power on a VM in VMware Workstation. But probably the most convenient way is to click the **Power on this virtual machine** link.
Do this now.

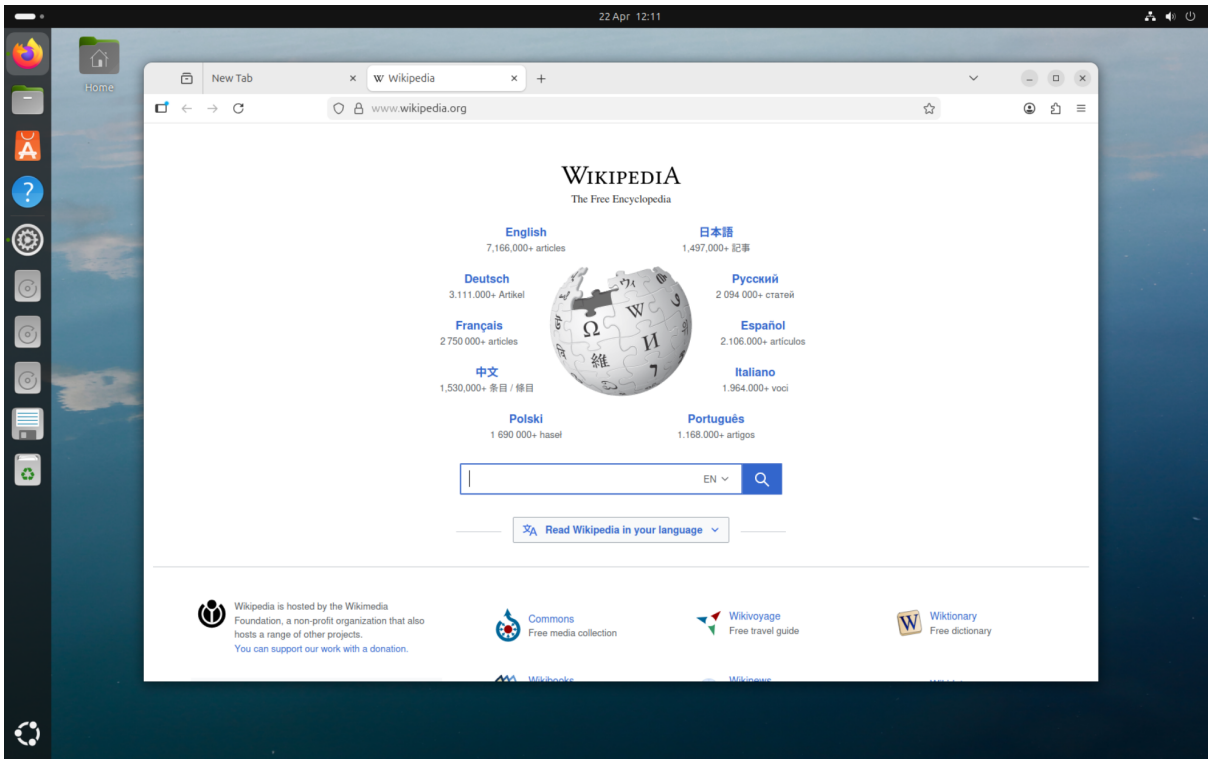
## 4. Updating and upgrading your Ubuntu Machine.

As with all operating systems, it's good practice to keep the software up-to-date. There is a **GUI** (graphical user interface) for this in Ubuntu but we need to practice with the **CLI** (command line interface).

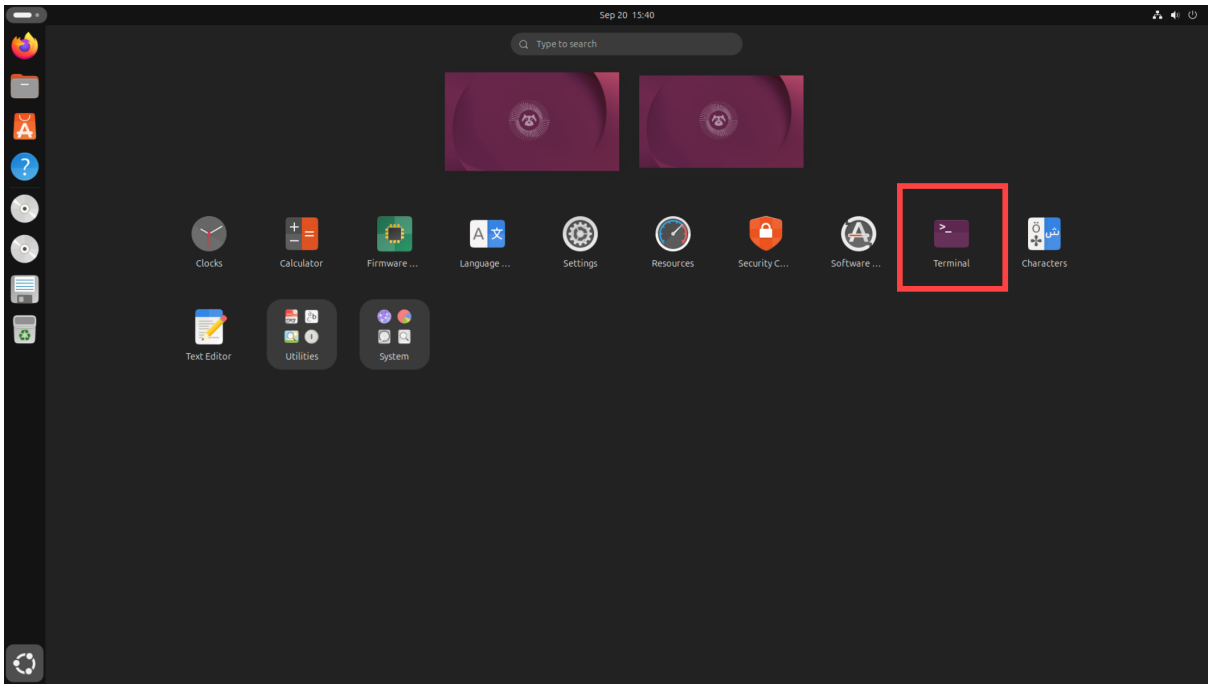
This is because not all Linux machines you will encounter will have a GUI – many Linux machines are created with only a CLI. GUI's use up resources unnecessarily, especially on Linux Servers.

Because updates and upgrades will require internet access, test to see if it's working on your VM:

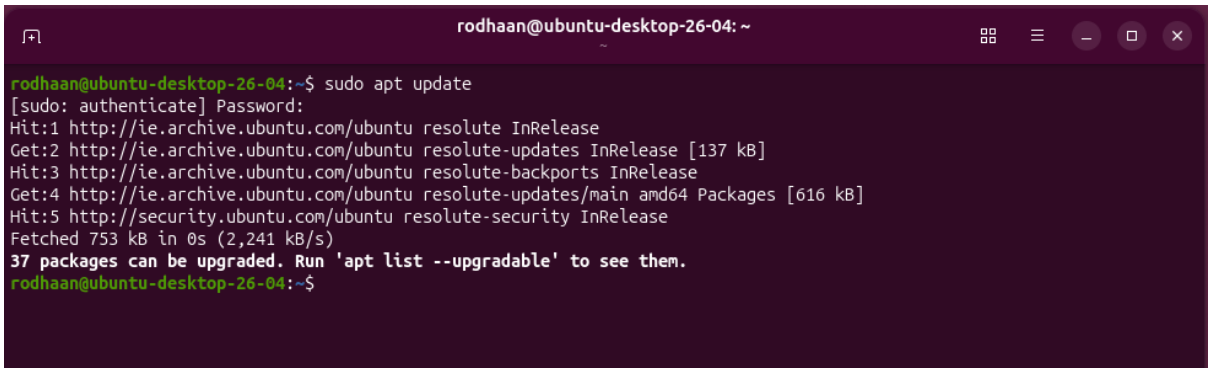
4.1. Open the Firefox web browser on your Ubuntu machine and navigate to any webpage. If this is successful, then your Ubuntu VM has internet access. Close the browser after checking.



4.2. Open the Terminal app.




4.3. Run `sudo apt update` and enter your password when prompted.



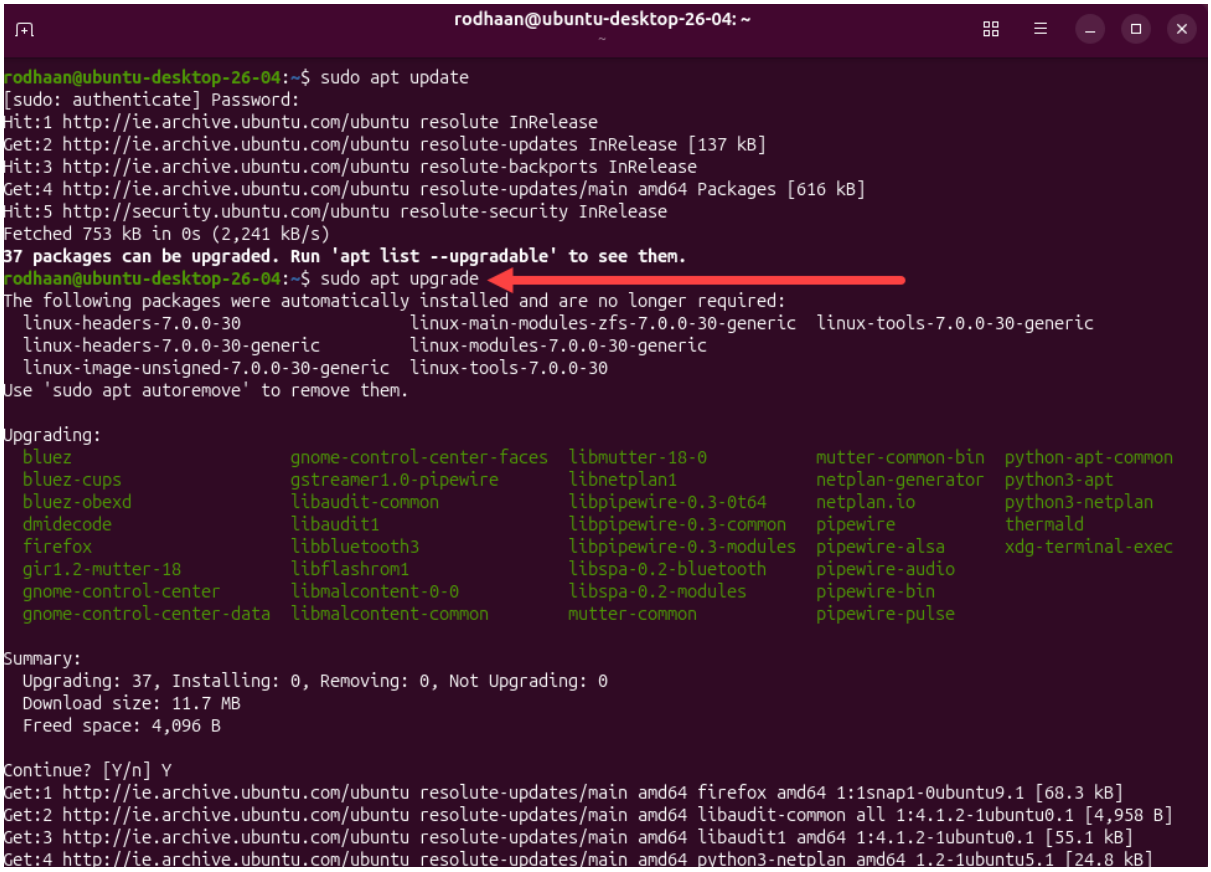

## What does the command mean?

`sudo` means “superuser do”. It runs the command with administrator privileges, usually after asking for your password.

`apt` stands for Advanced Package Tool. It manages software packages on Ubuntu, including installing, updating and removing software.

`update` tells APT to refresh its list of available packages and versions from the configured software repositories.

4.4. Run `sudo apt upgrade` and enter your password if prompted.




### What does the command mean?

`Upgrade` installs available updates for installed packages, using the package lists retrieved by `sudo apt update`.

**Notes:**

- As this is the first time your Ubuntu VM has been upgraded, it can take a few minutes to complete.
- You may also need to answer **Y** (yes) to some questions that appear during the upgrade.
- Running `sudo apt update` prior to installing any new packages is always good practice with Linux.

Typically, you will need to reboot your Linux VM after an `upgrade`. Ubuntu may prompt you to do so with a pop-up window. If it does, follow the instructions to reboot.

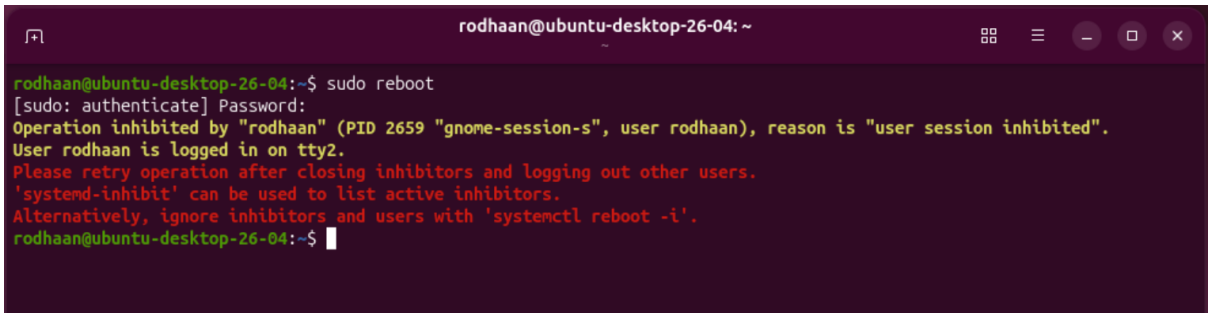
If there's no pop-up window, we can reboot from the CLI:

4.5. Run `sudo reboot`

4.6. Logon after rebooting the machine.

**Note:**

Reboot *inhibitors* are enabled on **Ubuntu 26.04 LTS** by default. You may see the message below:




Override the reboot inhibitors by running `sudo systemctl reboot -i`
The machine will reboot.

## 5. VMware Tools

By default, VMware enables the "*Easy Install*" feature (seen as the pale yellow bar underneath your VM during installation), when installing operating systems. "*Easy Install*" automatically installs VMware Tools on Ubuntu.

Having VMware Tools on your virtual machines is important because it provides the integration that makes a VM usable beyond a basic emulation. In other words, you get a much-improved VM experience. For example:

- Improved Graphics & Display Handling
- Mouse & Keyboard Integration
- Clipboard & Drag-and-Drop
- Shared Folders (HGFS)
- Improved Network Performance
- Time Synchronisation
- Graceful Shutdown & Power Operations
- Better Disk & Memory Performance
- Snapshot Consistency

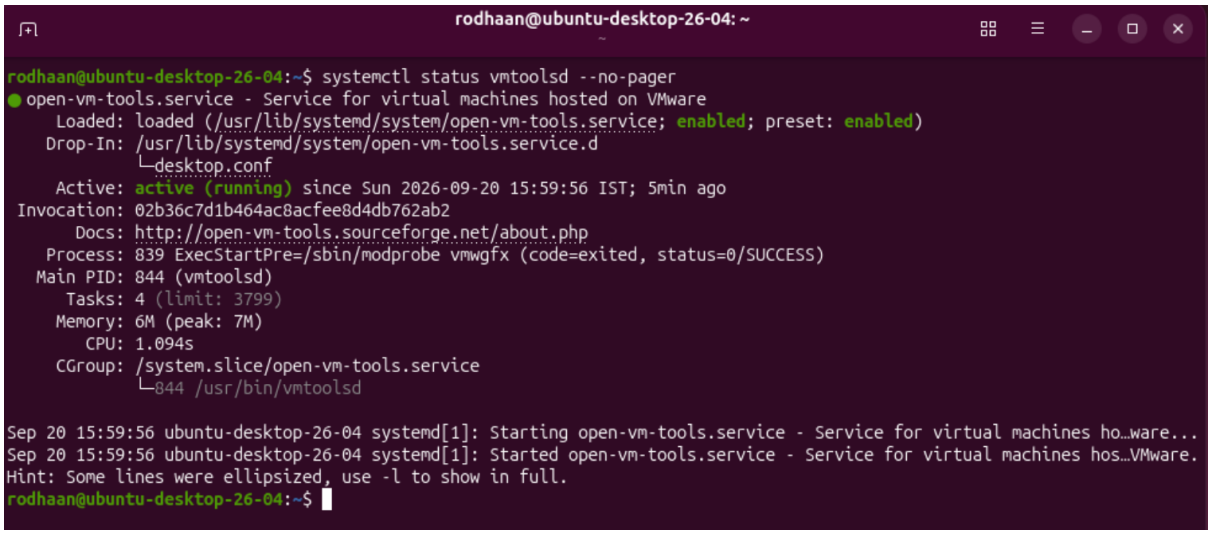
Since VM Tools should already be installed, all you need to do is confirm installation and correct operation.

5.1. Open the **Terminal** app.

5.2. Verify VMware Tools is installed, enabled, and active.
To do this:

Run `vmware-toolbox-cmd -v`
(displays the VMware Tools version)

Run `systemctl status vmtoolsd --no-pager`




Check that VMware Tools displays as `active (running)`.

## Important:

If VMware Tools is **not** installed on your Ubuntu machine, follow the instructions below to install them:

Run `sudo apt install -y open-vm-tools open-vm-tools-desktop`
(this installs VMware Tools)

Run `sudo systemctl enable --now vmtoolsd`
(this enables VMware Tools)

Run `sudo reboot`
(you need to reboot your Ubuntu VM after completing an upgrade or installing VMware Tools for the first time).

**The lab is now completed!**

Please ask you instructor to check your results.